# SEC P vs NP log 2026-05-18

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 21:12 UTC

## SEC P vs NP — daily log 2026-05-18

*31 cycles recorded*

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### Signed-Laplacian Holonomy Spectrum Bounds Tseitin Tree-Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete magnetic / signed Laplacian spectral theory of finite graphs — the $Z/2$ Bochner–Laplacian $L^\sigma$ for a line bundle on $G$ representing a cohomology class in $H^1(G; F_2)$, accessed through the gauge-invariant smallest eigenvalue $\lambda_{min}(L^\sigma)$ (Lieb–Loss 1993 on magnetic operators; Sunada 1994 on discrete magnetic Schrödinger; Mathai–Yates 2006; Atay–Liu 2014 and Lange–Liu–Peyerimhoff–Post 2015 on signed Cheeger inequalities). An arXiv search for 'magnetic Laplacian' AND ('Tseitin' OR 'resolution refutation' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers, none using gauge-invariant signed spectra as a CNF invariant; distinct from sandpile / 2-Sylow / Ihara-zeta attempts on Tseitin (those use cokernels, non-backtracking matrices, or critical-group orders, never the $F_2$ Bochner spectrum). × Tree-like Resolution / DPLL refutation size t*(T(G,ω)) for Tseitin XOR formulas T(G,ω) on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson–Wigderson 2001 / Alekhnovich–Razborov 2001 expansion regime), in the Cook–Reckhow–Krajíček bounded-arithmetic framework where ¬T(G,ω) is $\Sigma^b_0$ and $\log_2 t$* controls $V^0$-witnessing complexity.
- **Recorded**: 2026-05-18 00:33 UTC
- **Entry ID**: `2d264de23a0e`

#### Statement

For a connected 3-regular graph $G=(V,E)$ with odd charge $\omega:V\to F_2$ ($\Sigma_v\, \omega(v)\equiv 1 \bmod 2$), build a $Z/2$-gauge representative $\psi\in F_2^E$ by BFS from a fixed root $v_0$, setting each tree edge so that $\oplus_{e\ni v}\psi(e)=\omega(v)$ at every non-root $v$ (the unsatisfiability deficit lands at $v_0$); form the signed adjacency $A^\sigma$ with $\sigma(e)=(-1)^{\psi(e)}$ and the signed Laplacian $L^{\sigma=D-A}\sigma$, and define $\mu(G,\omega):=\lambda_{min}(L^\sigma)$, which is gauge-invariant under $\psi\mapsto\psi\oplus\delta f$ and equals 0 iff $\Sigma\omega$ is even. We conjecture an absolute constant $c>0$ such that for every unsatisfiable $T(G,\omega)$ on 3-regular $G$, $\log_2$ t*(T(G,ω)) ≥ c · μ(G,ω) · |V|, with pre-registered c=0.05. A single 3-regular instance (G,ω) with log_2 t*(T(G,ω)) < 0.05 · μ(G,ω) · |V| falsifies the conjecture.

#### Rationale

The Tseitin obstruction is exactly the $Z/2$ holonomy class $[\omega]\in H^1(G;F_2)$, and $\mu(G,\omega)$ is the spectral norm of this obstruction in the discrete Bochner–Laplacian sense — large ($\Theta(1)$) on expanders where holonomy is globally rigid, and small ($\Theta(1/|V|^2)$) on tree-like graphs where the obstruction concentrates locally. The mechanism funnels the Ben-Sasson–Wigderson expansion-based $\exp(\Omega(\text{boundary expansion}))$ lower bound through a single gauge-invariant scalar via the signed Cheeger inequality $h^\sigma(G)\ge\sqrt{2d\mu}$, making the standard expansion bound a corollary of a finer structural invariant.

The invariant is a property of the CNF instance (not of any truth-table), so it is shielded from natural proofs; it does not algebrize (signed Laplacian eigenvalues are not preserved under polynomial extension of the Boolean ring); and Tseitin lower bounds for tree-Resolution are not relativizing-style oracle facts.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 158, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 126, in run_trial
    psi = bfs_gauge(vertices, edges, 0)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 111, in bfs_gauge
    if adj[u][v] == 1 and not visited[v]:
       ^^^
NameError: name 'adj' is not defined
```

### Judge reasoning

Test crashed due to undefined variable ‘adj’ in BFS gauge calculation, preventing evaluation of conjecture validity | next: Fix the ‘adj’ variable definition in bfs_gauge function and rerun tests with multiple seeds to assess support fraction

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## Bonami-Beckner 4-2 Saturation of Clause-Sum Bounds DPLL Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bonami-Beckner hypercontractivity (Bonami 1970, Beckner 1975) — the degree-d 4-norm-vs-2-norm contraction ||P||_4 ≤ (q-1)^{d/2}||P||_2 — applied as a SYNTACTIC Fourier-analytic invariant of the centered clause-indicator polynomial Q_F of a CNF (a property of the formula’s clause/polarity structure, NOT of its truth table). An arXiv search ‘Bonami-Beckner hypercontractivity’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’) returns 0 direct hits, with <5 adjacent papers (Schoenebeck-style PC-degree, never DPLL search-tree depth via syntactic clause-sum 4-2 ratios). Distinct from the blacklisted Cross-Side Fourier attempt: that invariant was computed on the truth-table communication MATRIX (natural-proof shape); here Q_F lives on the clause-polarity STRUCTURE of F, so two semantically equivalent CNFs F1,F2 can have very different r(F), shielding the invariant from Razborov-Rudich constructivity. × DPLL / tree-like Resolution refutation depth d*(F) for unsatisfiable 3-CNFs, in the Ben-Sasson–Wigderson–Krajíček bounded-arithmetic regime where $\log_2$ d*(F) controls V^0-witnessing complexity of ¬F as Σ^b_0 and lower-bounds tree-Resolution width w*(F).
- **Recorded**: 2026-05-18 03:29 UTC
- **Entry ID**: `2b2653395609`

### Statement

For an unsatisfiable 3-CNF F with m clauses on n variables, let s_{i,v} ∈ {±1} be the polarity sign of variable v in clause C_i (s=+1 if v occurs negated, -1 if positive). Define the centered clause-falsification polynomial Q_F: {-1,+1}^n → R by Q_F(x) = Σ_{i=1}^m (∏*{v ∈ vars(C_i)} (1 + s*{i,v}x_v)/2 − 1/8), whose Fourier coefficient at T ⊆ [n] with |T| ∈ {1,2,3} is the closed-form sum Q̂*F(T) = (1/8) Σ*{i: T ⊆ vars(C_i)} ∏*{v ∈ T} s*{i,v} (computable directly from clause structure in O(m) time). Let r(F) := ||Q_F||_4 / ||Q_F||_2 ∈ [1, 3^{3/2}] be the Bonami-Beckner 4-2 ratio, and set G(F) := (3^{3/2} − r(F)) / 3^{3/2} ∈ [0, 1−3^{-3/2}]. Conjecture: there is an absolute constant c ≥ 0.05 such that for every unsatisfiable 3-CNF F, $\log_2(d^*(F) + 1) \geq c \cdot n \cdot G(F)$; a single F violating this inequality falsifies.

### Rationale

Hypercontractive saturation r(F) → 1 means Q_F has a heavy-tailed degree-3 spectrum: clause-violation behaves like a sharp witness function whose Fourier mass spreads thin, the regime characterizing expander-Tseitin and threshold-random 3-SAT where DPLL must explore exponentially many literal interactions. Conversely r(F) → 3^{3/2} (Khintchine-saturated) means Q_F is dominated by a few large coefficients, the regime of structurally simple unsat formulas that DPLL refutes shallowly. The bridge is novel because the Fourier analysis is performed on the CLAUSE STRUCTURE polynomial Q_F (syntactic), not on F's truth table or its communication matrix, sidestepping the natural-proofs largeness barrier that killed the Cross-Side Fourier attempt.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.18s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 141, in run_trial
    norm_4 = compute_norm_4(F, n)
             ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 81, in compute_norm_
    product *= (1 + s * x[v - 1]) / 2
                        ~^^^^^^^
IndexError: list index out of range
```

### Judge reasoning

The test crashed with an IndexError in compute_norm_4 before producing any data, so the pre-registered support condition across 480 instances could not be evaluated. | next: Fix the variable indexing bug in compute_norm_4 (likely a 1-based vs 0-based mismatch or variables exceeding n) and rerun the full 480-instance sweep.

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## Cycle-Type Mixing Defect of Barrington BP Bounds NC^1 Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Character-theoretic mixing on S_5 — the L^1 distance between the empirical cycle-type distribution of partial products of a deterministic word and the Haar cycle-type distribution π^* = (1,10,15,20,20,24,30)/120, in the Diaconis–Shahshahani 1981 / Lulov–Pak 2002 / Larsen–Shalev 2008 Plancherel-spectrum tradition. This is a STRUCTURAL invariant of the program (a property of the BP wiring, not of its truth table), and an arXiv search 'conjugacy

class mixing' AND ('branching program' OR 'Barrington' OR 'formula size') returns 0 direct hits with <5 adjacent papers, all on quantum-circuit Haar approximation. × NC^1 formula leaf size via Barrington 1989 width-5 permutation branching programs over S_5: every depth-d Boolean formula compiles into a length-4^d width-5 BP over S_5 whose terminal permutation is the identity iff f(x)=0 and the fixed 5-cycle σ_α iff f(x)=1. The target separation is NC^1 vs explicit P functions; the structural form of the invariant shields it from the Razborov–Rudich natural-proofs barrier (two formulas with identical truth tables can compile to BPs with very different MIX).
- **Recorded**: 2026-05-18 03:58 UTC
- **Entry ID**: `816a64f0e4e2`

### Statement

Let P=(I_1,…,I_L) be a width-5 BP over S_5 computing a Boolean f via Barrington's protocol, with I_t=(v_t,π_t$^{0,\pi_t}$1). For uniform X∈{0,1}^n let μ_t^P be the distribution on the 7 conjugacy classes of S_5 induced by Π_{s≤t}π_s^{X_{v_s}}, and define MIX(P):=(1/L)·Σ_{t=1}^L ‖μ_t$^{P-\pi}$*‖_1. Conjecture: for every non-constant f and every width-5 Barrington BP P computing f via the standard Ben-Or–Cleve commutator gadget, MIX(P)·L ≥ (1/8)·log_2 LeafSize(f), where LeafSize(f) is the minimum literal-leaf count of any AND/OR/NOT formula computing f. A single (P,f) with MIX(P)·L < (1/8)·log_2 LeafSize(f) refutes the conjecture.

### Rationale

Barrington-compiled BPs realize commutators of 5-cycles whose conjugacy-class trajectory is forced to spend most steps in non-trivial classes when the underlying formula has high algebraic depth; π^* assigns 119/120 mass off the identity class, so the deterministic word can only stay 'near uniform in cycle type' by genuinely commutator-rich computation. Linking MIX·L to log LeafSize makes mixing a syntactic complexity certificate that is bypassed by no truth-table proxy, addressing the ACC_SIPSER target of finding natural properties that are STRUCTURAL (not truth-table). The bridge to Lulov–Pak character bounds promises a path to ACC^0[m] via Barrington–Thérien compilation into solvable iterated wreath products without invoking arithmetization (avoiding ALGEBRIZATION).

### Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1107.1826v4] Conjugacy growth of finitely generated groups - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1512.02171v5] Right-jumps and pattern avoiding permutations - [1301.0379v2] The Parameterized Complexity of some Permutation Group Problems

### Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 108, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 85, in run_trial
    metric_value = barrington_protocol(formula, n)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 52, in barrington_pr
    product *= barrington_protocol(formula[bit], n)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 47, in barrington_pr
    if formula[bit] == 'x':
       ~~~~~~~^^^^^
IndexError: string index out of range
```

#### Judge reasoning

The test crashed with an IndexError in barrington_protocol before producing any RESULT line or data, so neither the support nor falsification condition could be evaluated. | next: Fix the recursive formula parser (handle empty subformulas and properly tokenize literals/operators) and re-run the 30-seed sweep.

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## NW-Design Restriction Dimension Separates Perm from Padded Det

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Nisan–Wigderson combinatorial designs (set systems with bounded pairwise intersection from PRG construction; Nisan–Wigderson 1994; Impagliazzo–Wigderson 1997) × Mulmuley–Sohoni GCT determinantal complexity dc(perm_n): orbit-closure separation of perm_n from padded det_m for $m \le n^{1.5}$ (Mignon–Ressayre 2004; Landsberg 2017)
- **Recorded**: 2026-05-18 04:25 UTC
- **Entry ID**: `0cac01ce9ab0`

#### Statement

Let $D=(S_1,\dots,S_M)$ be a Nisan–Wigderson design on the index set $[n]\times[n]$ of matrix-entry variables $x_{ij}$, with each $|S_t|=n$ and pairwise intersections $|S_s\cap S_t|\le k=\lceil\log_2 n\rceil$. For a homogeneous degree-n polynomial $f\in\mathbb{C}[x_{ij}]$*n, define the NW-restriction dimension* $\rho_D(f) := \dim_□ \mathrm{span}\{ f^{(t)} : t\in[M] \}$*, where* $f^{(t)}$ *is the polynomial obtained from f by setting x*$\{ij\}\mapsto 0$ for every $(i,j)\notin S_t$. CONJECTURE: there exist absolute constants $C_1,C_2>0$ with $C_2>4C_1\cdot\log_2 n / 1$ such that for every n and every M with $n \le M \le \lfloor n^{1.5}\rfloor$: (a) for every padded determinant $g(x)=□(x)^{n-m}\cdot\det_m(L(x))$ with $m\le\lfloor n^{1.5}\rfloor$, □ a linear form and L an m×m matrix of linear forms in the $x_{ij}$, $\rho_D(g) \le C_1\cdot m\cdot\log_2 n$; (b) $\rho_D(\mathrm{perm}_n) \ge C_2\cdot M$. A single (n,M,D,g) violating (a) or a single (n,M,D) violating (b) refutes the conjecture.

#### Rationale

NW designs probe disjoint-up-to-log-overlap windows of variables, so the M restrictions act like 'approximately independent samples' of the polynomial's syntactic content. A padded determinant compresses through only $O(m^2)$ hidden linear forms, so its NW-restrictions are forced to live in an O(m·polylog) dimensional subspace; the permanent has no such linear-form bottleneck, and its restrictions should be near-linearly-independent across NW slots. Because $\rho_D$ is an exp(n)-time syntactic invariant of the polynomial expansion (not of a truth table), it is shielded from Razborov–Rudich natural proofs, and the NW-design layer ties hardness-vs-randomness combinatorics to a GCT obstruction shape distinct from prior Specht/Mahonian/Eulerian attempts.

#### Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [2202.13016v1] Bounds on Determinantal Complexity of Two Types of Generalized Permanents - [1007.3804v4] Symmetric Determinantal Representation of Formulas and Weakly Skew Circuits - [0106076v3] A determinantal formula for the Hilbert series of one-sided ladder determinantal rings

### Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 163, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 130, in run_trial
    perm_rank = matrix_rank(perm_matrix)
                ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 33, in matrix_rank
    while pivot < n and matrix[pivot][col] == 0:
                        ~~~~~~~~~~~~~^^^^^
IndexError: list index out of range
```

### Judge reasoning

The test crashed with an IndexError in matrix_rank before any trials completed, so no data was produced to evaluate the pre-registered support or falsification conditions. | next: Fix the matrix_rank indexing bug (likely a row/column dimension mismatch when building perm_matrix) and rerun the full 270-trial sweep.

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## Coxeter Inversion Spread of Barrington BPs Bounds NC^1 Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coxeter-group combinatorics of S_5 — the inversion-count / Coxeter length □_C: S_5 → {0,1,…,10} relative to the simple-transposition generators s_i=(i,i+1), in the Bjorner–Brenti / Humphreys finite-Coxeter-systems tradition (reflection length, weak order, Eulerian-statistics). This corner of finite Coxeter combinatorics is essentially absent from NC^1 / Barrington literature: an arXiv search ‘Coxeter length’ AND (‘Barrington’ OR ‘branching program’ OR ‘NC^1’ OR ‘formula size’) returns 0 direct hits and <5 adjacent papers, none using inversion-count statistics on BP partial-product traces. Distinct from the blacklisted character-mixing attempt: that invariant was a Plancherel/cycle-type Fourier functional, while □_C is a NON-ring, geometric word-length on S_5 (so it does not extend to polynomial relaxations and cannot algebrize). × NC^1 formula leaf size via Barrington 1989 width-5 permutation BPs over S_5: every depth-d Boolean formula compiles into a length-4^d width-5 BP whose terminal permutation is e iff f(x)=0 and a fixed 5-cycle σ iff f(x)=1. The structural form of the invariant (a property of the BP wiring, not of the truth table) shields it from the Razborov–Rudich natural-proofs barrier, since two BPs with identical truth tables can yield different σ² values.
- **Recorded**: 2026-05-18 04:47 UTC
- **Entry ID**: `66b589aa0f3d`

### Statement

Let B be a width-5 Barrington BP of length L on n Boolean inputs, with layers (i_k, g_k^0, g_k^1)*{k=1..L}; for x ∈ {0,1}^n let π_k(x) := g_1^{x*{i_1}} · g_2^{x_{i_2}} · … · g_k^{x_{i_k}} ∈ S_5 and let □_C(π) be the inversion count of π. Define σ²(B) := E_{x~U({0,1}^n)} [ (1/L)·Σ_{k=1}^L (□_C(π_k(x)) − μ_x)² ], where μ_x := (1/L)·Σ_k □_C(π_k(x)). Conjecture: (i) for every BP B obtained by Barrington-compiling a depth-d AND/OR/NOT formula via the canonical Ben-Or–Cleve construction (so L = 4^d), σ²(B) ≤ C_1 · d for an absolute constant C_1 ≤ 4; (ii) for every non-constant width-5 BP B (Barrington-compiled or arbitrary), σ²(B) ≥ c_2 > 0 with c_2 ≥ 1/8. A single BP violating either bound refutes.

### Rationale

Coxeter length is a quantitative geometric measure of how far the partial-product walk wanders inside the Cayley graph Cay(S_5, simple transpositions); Barrington-compiled BPs traverse highly structured commutator-stamped excursions whose inversion-trace has tightly-controlled fluctuations governed by the recursion depth, while any non-trivial BP must achieve at least one □_C > 0 step and thus a nonzero variance. If true, $\sigma^2$ provides a depth-sensitive, truth-table-blind certificate that lower-bounds the formula depth realising a given BP — a structural NC^1 separator immune to natural-proofs and algebrization, since □_C is not a polynomial functional and does not extend to any commutative-ring relaxation of the Boolean cube.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 84, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 55, in run_trial
    layers = barrington_compile(int(n**(1/4)))
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 30, in barrington_co
    layers.append(permutation_multiply(sigma, sigma))
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 24, in permutation_m
    return tuple(p1[p2[i]] for i in range(len(p1)))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 24, in <genexpr>
    return tuple(p1[p2[i]] for i in range(len(p1)))
                 ~~^^^^^^^
IndexError: tuple index out of range
```

### Judge reasoning

The test crashed with an IndexError in permutation_multiply before any $\sigma^2(B)$ data was produced, so neither the support nor the falsification condition can be evaluated. | next: Fix the Barrington compiler bug (permutation_multiply receiving a malformed tuple from barrington_compile) and rerun the 30×4 + 2 probe protocol.

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## Birkhoff Spectral Gap of Gate Co-occurrence Bounds AC^0 PARITY Correlation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Birkhoff polytope / doubly-stochastic spectral theory (Sinkhorn 1964; Linial–Samorodnitsky–Wigderson 2000; Forbes–Saxena 2018 on permanental approximations) applied as a STRUCTURAL invariant of the gate-pair input co-occurrence matrix of a Boolean circuit. The Sym(n)-equivariant Sinkhorn scaling and its λ_2 spectral gap are a non-ring convex-analytic invariant (the scaling map uses divisions and row/column normalisations that do not extend to F_q, so it cannot algebrize). An arXiv search for 'Sinkhorn scaling' AND

(‘AC^0’ OR ‘switching lemma’ OR ‘circuit complexity’) returns 0 direct hits and <5 adjacent papers, all on permanent approximation in algebraic complexity, never on AC^0 PARITY correlation. × AC^0[d] circuit correlation with PARITY_n in the Håstad 1986 switching-lemma / Razborov–Smolensky regime, in the Cook–Reckhow–Krajíček bounded-arithmetic framework where AC^0 PARITY size lower bounds correspond to V^0 ￿ V^0(⊕) separations. The invariant is a STRUCTURAL property of the circuit DAG (not of its truth table), shielding it from the Razborov–Rudich natural-proofs barrier; it is Sym(n)-equivariant under the natural action permuting input variables, fulfilling the focus’s ‘natural with respect to a non-trivial group action’ requirement.

- **Recorded**: 2026-05-18 05:16 UTC
- **Entry ID**: `14ed7020b118`

### Statement

For an AC^0 circuit C on n inputs of depth d and size s, let I(g)⊆[n] be the input cone of gate g (variables reaching g in C). Define A(C)∈R$^{n\times n}$*{≥0} by A[i,j] = #{gates g : {x_i,x_j} ⊆ I(g)}, set Ã = A + 10^{-3}·J, and let D(C) be the doubly-s*
satisfies κ(C) ≤ exp(− c · n · g(C) / d). A single AC^0 circuit C with κ(C) > exp(−c·n·g(C)/d) (for c = 1/8) refutes the conjecture.

### Rationale

Hastad’s switching lemma kills AC^0 correlation with PARITY via random restrictions that spread mass evenly across input coordinates; the Sinkhorn-doubly-stochastic projection captures the SAME spreading geometrically — g(C) measures how ‘mixed’ the gate-pair co-occurrence matrix is under symmetric scaling, and large g(C) forces no input pair to dominate the circuit’s structure. Because D(C) transforms equivariantly under Sym(n) permuting inputs, g(C) is a representation-theoretic invariant of the circuit’s wiring, not of its truth table — so it cannot be a Razborov–Rudich natural property. The construction is purely convex-analytic (Sinkhorn uses divisions, not ring operations) and so does not algebrize under polynomial extensions of the Boolean ring, sidestepping the AVOID examples (tropical homotopy stability, tropical weight accumulation) that hit ALGEBRIZATION.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `0`, elapsed: `5.54s`


```
jecture_holds': True, 'counterexample': '', 'n': 16, 'd': 3, 's': 48, 'g': -
0.7359093566514678, 'kappa': 0.0, 'bound': 1.6333060917151618}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8870418116493577, 'kappa': 0.0, 'bound': 1.8064456904906498}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8660207193645115, 'kappa': 0.0, 'bound': 1.5418921941310963}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.805622159349614, 'kappa': 0.0, 'bound': 1.4960242355796014}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.9261501129555745, 'kappa': 0.0, 'bound': 1.5889525991259912}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8006633967206949, 'kappa': 0.0, 'bound': 1.4923196155245912}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.7322580901888673, 'kappa': 0.0, 'bound': 2.07977162120796}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': '
```

### Judge reasoning

Every trial reports $\kappa=0.0$ exactly, so the inequality holds vacuously; additionally g(C) is frequently negative, which is impossible for a genuine doubly-stochastic matrix and indicates a Sinkhorn/eigenvalue bug. The pre-registered criterion ($\geq$2000 trials, Spearman $\rho \geq 0.3$) is not demonstrated, and the regime tested ($n \leq 16$, tiny random circuits) never approximates PARITY. | next: Fix the Sinkhorn scaling (verify convergence and that $|\lambda\_2(D)| \leq 1$), then test on circuits designed to have nontrivial PARI

---

## Mansour L1 Fourier Mass of Clause-Falsification Polynomial Bounds DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier analysis of Boolean functions — specifically the Linial–Mansour–Nisan / Mansour L^1 Fourier mass L_1(p) := Σ_S |p̂(S)| of low-degree pseudo-Boolean polynomials in the Bonami–Beckner–Mansour tradition; distinct from the 4-norm/2-norm Bonami ratio (blacklisted) and from the cross-side symmetric-difference mass (blacklisted), since L_1 is the sum-of-absolute-coefficients functional underlying Mansour's DNF conjecture and the LMN approximation theorem. An arXiv search for 'Mansour L1 Fourier mass' AND ('Resolution refutation' OR 'DPLL' OR 'tree-Resolution') returns 0 direct hits and <5 adjacent papers, none using L_1 of the clause-count polynomial as a CNF-structural invariant. × Tree-like Resolution / DPLL refutation size t*(F) for unsatisfiable 3-CNFs F in the Ben-Sasson–Wigderson / Krajíček bounded-arithmetic regime, where ¬F is Σ^b_0 and log_2 t*(F) lower-bounds V^0-witnessing complexity; structural shielding from Razborov–Rudich is provided because L_1 depends on the clause LIST (multiplicities, polarities), so two unsat CNFs with identical satisfiability functions (both ≡ FALSE) can have very different L_1.
- **Recorded**: 2026-05-18 05:56 UTC
- **Entry ID**: `9076d5e87956`

### Statement

For an unsatisfiable 3-CNF F = {C_1,...,C_m} on n Boolean variables, let p_F: {-1,+1}^n → Z, p_F(x) := #{j : x falsifies C_j} be the clause-falsification count polynomial, expand it in the Walsh basis p_F(x) = Σ_S p̂*F(S) χ_S(x), and define the structural Fourier L^1 mass L_1(F) := Σ*{1 ≤ |S| ≤ 3} |p̂_F(S)|. We conjecture that for every unsatisfiable 3-CNF F there is an absolute constant c ∈ [0.25, 3.0] such that L_1(F) / √(m+1) ≤ c · √( log_2( t*(F) + 1 ) + 1 ), where t*(F) is the DPLL tree size under the most-frequent-variable branching heuristic; a single unsat F violating the inequality refutes the conjecture.

### Rationale

Mansour's conjecture (DNF L_1 ≤ s^{O(loglog s)}) and the LMN low-degree concentration theorem tie Boolean-function L^1 Fourier mass to small-formula computability; transposing this to CNF refutations predicts that 'Fourier-spread' clause sets (large L_1 per √m) are forced to have correspondingly deep search trees, since each Walsh monomial in p_F encodes a clause-polarity correlation that any DPLL path must locally certify. The Tseitin family is a sharp test bed: every XOR=ω(v) constraint contributes Fourier mass concentrated exactly at one size-3 monomial of magnitude 1/2 (lower levels cancel by parity), giving L_1(Tseitin)/√m = Θ(√V), parallel to the Ω(V) Ben-Sasson–Wigderson width lower bound. The invariant is structural (clause-list-dependent), not algebrizing (uses |·| not ring extensions), and not relativizing.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.03s


```
nces_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conject
RESULT: INCONCLUSIVE reason=no_valid_instances
```


### Judge reasoning

The test generated zero valid instances across all trials (RESULT: INCONCLUSIVE reason=no_valid_instances), so the pre-registered support condition of >=200 instances was not met and no empirical evaluation of L_1(F) or t*(F) actually occurred. | next: Fix the instance generator to reliably produce unsatisfiable 3-CNFs from the Tseitin, random-density-5, and trivial-unsat families, verify nonzero instances_tested, and re-run with the 30-seed protocol.

---

## Chang Spectrum Dimension of Wire-Source DFS Labels Bounds ACC^0[2] for MOD_3

- **Verdict**: INCONCLUSIVE
- **Bridge**: Additive combinatorics — Chang's spectral lemma (Chang 2002; Bourgain 2008; Sanders 2019; Green–Ruzsa) bounding the size of the η-spectrum Spec_η(A)⊂Ẑ/p of a set A⊂Z/p of density α by O(log(1/α)/η²); applied as a non-ring, Fourier-analytic STRUCTURAL invariant of the circuit DAG (wire-source integer multisets under canonical DFS-from-output labeling), NOT of the truth table. Distinct from the avoided Plünnecke top-Fourier attempt, which used a TRUTH-TABLE Fourier functional; here the Fourier transform is over wire LABELS in Z/p (gauge fixed by a canonical DFS), so two semantically equivalent circuits yield different χ values, shielding from Razborov–Rudich. The invariant uses |·| and FFT (does not extend to polynomial F_q relaxations), so it does not algebrize per Aaronson–Wigderson. An arXiv search 'Chang spectral lemma' AND ('ACC^0' OR 'circuit complexity' OR 'Williams algorithmic method') returns 0 direct hits and <5 adjacent papers (all on Bohr-set Roth bounds in pure additive combinatorics). × ACC^0[2] circuit size for MOD_q on n inputs with q coprime to 2 (focus q=3), in the Williams 2011 / Murray–Williams 2018 algorithmic-method regime targeting NEXP ⊄ ACC^0 and explicit-function ACC^0 lower bounds (Sipser-style candidates).
- **Recorded**: 2026-05-18 06:21 UTC
- **Entry ID**: 29a5335efcbb

### Statement

Let C be an ACC^0[2] circuit of size s on n inputs, with gates topologically ordered g_1,…,g_s by a canonical DFS-from-output labeling (children visited in lexicographic gate-type then wire-index order). Let p be the smallest prime with p ≥ s+2. For each MOD_2 gate g of fan-in k ≥ 4 with input gate-ids a_1,…,a_k ∈ [s], define the spectrum dimension χ(g) := |{ξ ∈ {1,…,p−1} : |Σ_{j=1}^k e^{2πi a_j ξ/p}| ≥ k/2}|, and χ(C) := (max_g χ(g))/log_2(p+1) (taking 0 if C has no MOD_2 gate with fan-in ≥ 4). Conjecture: (i) for every ACC^0[2] circuit C of size s, χ(C) ≤ C_1·log_2(s+1) with C_1 = 8; and

(ii) every ACC^0[2] circuit C computing MOD_3_n exactly satisfies $\chi(C) \geq C_2 \cdot n/(\log_2(n+2))^2$ with $C_2 = 1/4$. A single random ACC^0[2] circuit violating (i) or a single MOD_3-computing ACC^0[2] circuit violating (ii) refutes the conjecture.

### Rationale

Chang's theorem says that the η-spectrum of a random or generic subset of Z/p is small (O(log p)), so for typical circuits the structural invariant χ(C) should saturate the upper bound (i). Computing MOD_3, however, requires the wire-source MOD_2 gates to concentrate Fourier mass at the cube-root-of-unity-like frequencies of Z/p (because PARITY-composition pathways must reconstruct triadic structure), which forces a Bohr-set-shaped wire-label pattern with large spectrum — giving the lower bound (ii). Because χ depends on the canonical DFS LABELS of the wires (a property of the circuit DAG, NOT of its truth table), the invariant is shielded from Razborov–Rudich, and because it relies on complex-exponential sums (not on ring operations extendable to polynomial relaxations), it is shielded from Aaronson–Wigderson algebrization.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e94115ad.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e94115ad.py", line 51, in run_trial
    circuit = generate_random_acc02_circuit(n, s)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e94115ad.py", line 22, in generate_rand
    fanin = random.randint(2, min(len(gates) + 1, math.floor(math.log2(s)) + 3))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(2, 2)
```

### Judge reasoning

The test crashed with a ValueError in generate_random_acc02_circuit before any trials completed, so no data was produced to evaluate either bound (i) or (ii). | next: Fix the fan-in sampling bug (guard min(len(gates)+1, floor(log2(s))+3) ≥ 2, e.g., skip MOD gate creation until enough predecessor gates exist) and rerun the 360 random instances plus the MOD_3 enumeration.

---

## Bourgain Noise-Sensitivity Mass of Clause-Falsification Polynomial Lower-Bounds DPLL Size

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Bourgain noise-sensitivity / sharp-threshold Fourier analysis — the high-frequency Fourier functional $\mathrm{NSM}(f) = \Sigma_S (1-\rho^{|S|})\cdot\hat{f}(S)^2$ as a structural invariant, in the Bourgain 1999 / Kahn–Kalai–Linial 1988 / Friedgut–Kalai 1996 / Mossel–O'Donnell–Oleszkiewicz 2010 tradition. Distinct from Mansour-L1 ($\Sigma_S |\hat{f}(S)|$, blacklisted), the Bonami 4-norm/2-norm ratio (blacklisted), and the cross-side symmetric-difference mass (blacklisted, NATURAL_PROOFS-hit on a truth-table Fourier object): NSM is a quadratic, level-weighted Fourier mass that captures Bourgain-style 'sensitivity at scale 1/n' rather than absolute coefficient sums. The functional uses absolute values and a level-indexed weighting $\Sigma_S (1-\rho^{|S|})$ that does not extend to a polynomial $F_q$ relaxation, so the bridge does not algebrize per Aaronson–Wigderson. An arXiv search for 'Bourgain noise sensitivity' AND ('DPLL' OR 'tree-Resolution' OR 'CNF refutation') returns 0 direct hits and <5 adjacent papers (none using NSM on the syntactic clause-falsification polynomial as a CNF invariant). × Tree-like Resolution / DPLL refutation size t*(F) for unsatisfiable 3-CNFs F in the Ben-Sasson–Wigderson / Krajíček bounded-arithmetic regime where ¬F is* $\Sigma^b_0$ *and* $\log_2 t$(F) lower-bounds $V^0$-witnessing complexity. Structural shielding from Razborov–Rudich is provided because NSM(F) is a functional of the clause LIST (multiplicities, polarities) — two unsat CNFs with identical satisfiability functions (both ≡ FALSE) can have very different NSM(F) — so the invariant is not a poly-time-computable property of truth tables.
- **Recorded**: 2026-05-18 06:47 UTC
- **Entry ID**: `2f957f34a9d5`

### Statement

For every unsatisfiable 3-CNF F on $n \geq 10$ variables with m clauses, define $\psi_F: \{-1,+1\}^n \to [0,1]$ by $\psi_F(x) = (1/m)\cdot\#\{C \in F : C \text{ is falsified by } x\}$, with Fourier expansion $\psi_F = \Sigma_S\, \hat{\psi}$*F(S)·χ_S under uniform measure. Set* $\rho = 1 - 1/n$ *and define* $\mathrm{NSM}(F) := (\mathrm{Var}(\psi_F) - \mathrm{Stab}_\rho(\psi_F)) / \mathrm{Var}(\psi_F) =$ $\Sigma\{S\neq\emptyset\}(1-\rho^{|S|})\cdot\hat{\psi}$*F(S)² / Σ*$\{S\neq\emptyset\}\ \hat{\psi}_F(S)^2 \in [0,1]$. Conjecture: there is an absolute constant $c \geq 1/16$ such that $\log_2 t$*(F) ≥ c · n · NSM(F), where t*(F) is the size of the canonical tree-Resolution refutation produced by a fixed deterministic DPLL with unit propagation and lexicographic branching; a single unsat 3-CNF violating this inequality refutes the conjecture.

### Rationale

Bourgain's noise-sensitivity machinery isolates 'high-frequency' Fourier mass of structural polynomials; for an unsat CNF, large NSM(F) means $\psi_F$'s Fourier energy sits at levels $\Omega(n)$, i.e. the falsification pattern depends on $\Theta(n)$-tuples of variables rather than a small junta. Because DPLL's branching corresponds to restriction operators on $\psi_F$ and a small refutation tree exhibits a low-junta restriction-tree for $\psi_F$ (each leaf restricts $\psi_F$ to a single falsified clause), high noise-sensitivity should force exponentially many leaves — paralleling how KKL/Bourgain bound junta-approximation by noise sensitivity. The invariant is syntactic (clause-list dependent), quadratic in Fourier coefficients with a non-ring level-weighting, and computable in $2^n \cdot n$ time, so it sidesteps Natural Proofs, Algebrization, and Relativization simultaneously.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `0`, elapsed: `0.07s`

```
No unsat instances found'}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
```

```
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False
RESULT: INCONCLUSIVE reason=insufficient_support
```


#### Judge reasoning

The test generator produced zero unsatisfiable 3-CNF instances across all trials, and the RESULT line explicitly reports INCONCLUSIVE due to insufficient support. The pre-registered threshold of ≥240 accepted instances was not even approached (0 tested). | next: Replace the random-generation pipeline with a guaranteed-unsat construction (e.g., random 3-CNF at α≈4.6 filtered by a SAT solver, or structured families like pigeonhole/Tseitin encoded as 3-CNF) to ensure ≥240 accepted instances before

---

## Generated Algebra Dimension of Layer Operators Bounds IP_2 Read-Twice BP

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Operator algebra / Burnside-Schur-Wedderburn theory — the unital R-subalgebra A(P) ⊆ R^{w×w} generated by the layer transition matrices {T_i^{(b)}} of a branching program P, with dim A(P) computed by iterative span closure (start from identity and all T_i^{(b)} flattened to R^{w^2}; multiply each existing basis element by each layer operator, add the product to the span via Gauss-Jordan, iterate until no new dimension is found). Distinct from prior free-probability attempts (S-transform, Laplacian R-transform, free cumulants on Max-Cut) which targeted communication / SDP gaps; here the algebra invariant is a STRUCTURAL property of the BP wiring (not its truth table — two BPs computing the same function can have very different dim A), shielding from Razborov-Rudich. The closure uses real linear-algebra rank operations (not ring identities over F_q), shielding from Aaronson-Wigderson algebrization. An arXiv search 'generated matrix algebra' AND ('branching program' OR 'read-twice' OR 'Borodin Razborov Smolensky') returns 0 direct hits with <5 adjacent papers (all on Barrington permutation BPs over a fixed simple group algebra, never on the generated algebra of arbitrary read-twice transitions). × Read-twice oblivious BP size s(P) for IP_2(x,y) = XOR_k (x_k AND y_k) on 2n inputs under the adversarial variable order x_1,…,x_n followed by y_1,…,y_n (Borodin-Razborov-Smolensky 1993 regime); instantiating the focus's 'distinguishing tensor' formalism: find ρ(P) with ρ(P) = O(log size(P)) but ρ(P_IP) = Ω(n) for any P computing IP_2.
- **Recorded**: 2026-05-18 07:28 UTC
- **Entry ID**: `59a86c7a7d5c`

#### Statement

Let P be a read-twice oblivious BP on 2n inputs with width w, length L, and 2L layer transition matrices {T_i^{(b)} ∈ {0,1}^{w×w} : i ∈ [L], b ∈ {0,1}} where each row of T_i^{(b)} has at most one 1. Let A(P) ⊆ R^{w×w} be the unital R-subalgebra generated by the identity and the {T_i^{(b)}}, and define ρ(P) := log_2(dim_R A(P) + 1). Conjecture: (i) for every read-twice oblivious BP P, ρ(P) ≤ 2·log_2(w + 1) (trivially since dim A(P) ≤ w^2 ≤ size(P)^2); (ii) for every read-twice oblivious BP P computing IP_2 under the adversarial input order, ρ(P) ≥ n − 2. A single read-twice oblivious BP P with f_P = IP_2 (adversarial order) and ρ(P) < n − 2 refutes (ii); a single BP violating (i) refutes (i).

**Rationale**

A(P) encodes EVERY product of layer transitions, i.e. all reachable state-permuters of the BP's syntactic dynamical system — a relational/categorical span, not a single truth-table number, so it dodges the textbook Razborov-Rudich trap and the recent calibration warnings against truth-table single-number invariants. By the BRS rectangle argument any read-once BP for IP_2 in adversarial order must distinguish 2^n x-prefixes at the midpoint cut; we conjecture this distinguishing power forces a Schur-Wedderburn-large generated algebra even with two reads per variable, because the y-phase must non-trivially recombine all 2^n prefix-states into a single parity bit. The span-closure construction uses real rank operations only, so it does not extend to a polynomial F_q relaxation and is shielded from Aaronson-Wigderson algebrization.

**Novelty**

- Judge: `NOVEL` over 0 arXiv hits

**Empirical Test**

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 158, in <module>
    main()
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 140, in main
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 123, in run_trial
    rho = compute_rho(P, var_positions, w)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 110, in compute_rho
    dimension = compute_dimension(basis)
                ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 60, in compute_dimen
    basis = gaussian_elimination(basis)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 52, in gaussian_elim
    matrix[k][j] -= factor * matrix[i][j]
    ~~~~~~~~~^^^
IndexError: list index out of range
```

**Judge reasoning**

The test crashed with an IndexError inside gaussian_elimination before producing any RESULT line or data, so neither the support nor falsification condition can be evaluated. | next: Fix the Gaussian elimination routine (ensure consistent row/column dimensions and bounds checks) and rerun the 360 random + canonical/friendly IP_2 trials.

---

## Szegedy Quantum-Walk Phase Gap Lower-Bounds Tseitin Tree-Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete-time Szegedy / Grover quantum walks on finite graphs (Szegedy 2004 'Quantum speed-up of Markov chain based algorithms'; Magniez–Nayak–Roland–Santha 2011 'Search via quantum walk'; Krovi–Magniez–Ozols–Roland 2016) — the phase gap Δ_Q(G) of the Szegedy walk operator W=(2Π_B−I)(2Π_A−I) acting on the edge-orientation space C^{2m}, accessed through the Szegedy correspondence Δ_Q = 2·arcsin(√gap(P_G)) where P_G is the

simple-random-walk transition matrix. This corner of quantum-walk operator theory is essentially absent from proof complexity: an arXiv search 'Szegedy quantum walk' AND ('Tseitin' OR 'resolution refutation' OR 'DPLL' OR 'proof complexity') returns 0 direct hits, with <5 adjacent papers (all on quantum mixing-time speedups, never on CNF-refutation lower bounds). Distinct from the blacklisted magnetic-Laplacian holonomy attempt: that invariant was the smallest gauge-invariant eigenvalue of a signed Bochner operator on 0-forms, while Δ_Q is a phase-gap on the EDGE-ORIENTATION (line-graph-like) space, computed from singular values of the discriminant operator rather than gauge-fixed signed spectra. × Tree-like Resolution / DPLL refutation size t*(T(G,ω)) for Tseitin XOR formulas T(G,ω) on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson–Wigderson 2001 / Alekhnovich–Razborov 2001 expansion regime), in the Cook–Reckhow–Krajíček bounded-arithmetic framework where ¬T(G,ω) is Σ^b_0 and log_2 t* lower-bounds V^0-witnessing complexity.
- **Recorded**: 2026-05-18 08:08 UTC
- **Entry ID**: `97906ec529b6`

### Statement

For a connected 3-regular graph G on n vertices with odd charge ω, define ν(G) := ⌊n · Δ_Q(G)⌋ where Δ_Q(G) = 2·arcsin(√(1−|λ_2(P_G)|)) is the Szegedy quantum-walk phase gap on G's edge-orientation space and λ_2(P_G) is the second-largest-magnitude eigenvalue of the simple-random-walk matrix P_G = D^{-1}A. Then there exist absolute constants c_1, c_2 > 0 such that (i) every 3-regular G with min balanced edge-bisection ≤ √n satisfies ν(G) ≤ c_1, and (ii) for every odd ω, t*(T(G,ω)) ≥ 2^(c_2·ν(G)). A single 3-regular G with odd ω and t*(T(G,ω)) < 2^(ν(G)/8) refutes (ii); a single G with bisection ≤ √n and ν(G) > 16 refutes (i).

### Rationale

The Szegedy walk's phase gap encodes a topological obstruction to coherent superposition mixing across edge ORIENTATIONS — exactly the structure that Tseitin XOR formulas force a Resolution refutation to track via the F_2-cycle-space mechanism (Urquhart–BSW). Because Δ_Q is built on the line-graph edge space rather than the vertex Laplacian, it captures orientation-sensitive bottlenecks invisible to plain vertex spectral gap, while remaining poly-time computable via one symmetric eigendecomposition. The invariant is shielded from Razborov–Rudich (structural property of G, not of any truth table) and from Aaronson–Wigderson (the arcsin/√ operations do not extend to a polynomial F_q relaxation).

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_086fec4c.py", line 225, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_086fec4c.py", line 198, in run_trial
    lambda_2 = compute_second_largest_eigenvalue(P)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_086fec4c.py", line 138, in compute_seco
    eigenvalues = matrix_eigenvalues(P)
                  ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_086fec4c.py", line 91, in matrix_eigenv
    raise ValueError("Matrix too large for this implementation")
```

```
ValueError: Matrix too large for this implementation
```

### Judge reasoning

The test crashed with a ValueError ('Matrix too large for this implementation') in the eigenvalue computation before any of the 270 instances could be evaluated, so neither support nor falsification conditions were tested. | next: Replace the custom matrix_eigenvalues routine with numpy.linalg.eig or scipy.sparse.linalg.eigs to handle the 2n×2n matrices for n∈{12,16,20}, then rerun the 270-instance protocol.

---

## Halász L^2 Spectrum Discrepancy Lower-Bounds Sign-Matrix Rigidity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Erdős–Turán / Halász L^2 discrepancy of empirical measures on [0,1] (Halász 1981 'On the boundedness of digital nets discrepancy'; Drmota–Tichy 1997 ch.1; Vaaler 1985 majorant–minorant; Niederreiter–Pillichshammer QMC tradition) — a non-ring, integral-of-CDF-deviation functional that uses sup/abs/integration and does NOT extend to polynomial relaxations over F_q (Aaronson–Wigderson algebrization-safe). Applied here as the L^2 deviation of the rescaled empirical SQUARED-singular-value CDF of a sign matrix against the maximum-entropy (uniform-on-spectrum) reference. An arXiv search 'Halász discrepancy' AND ('matrix rigidity' OR 'Valiant rigidity') returns 0 direct hits; <5 adjacent papers on QMC discrepancy vs spectral concentration of random ±1 ensembles, none targeting rigidity. Distinct from blacklisted Bourgain–Tzafriri sub-column dispersion (column-subset operator norms) and signed-support L_inf discrepancy (Beck–Fiala on incidence rows). × Matrix rigidity R_M(r) for sign matrices M∈{±1}^{N×N} in the Valiant 1977 / Friedman 1993 / Razborov 1989 / Lokam 2009 regime, accessed through the Friedman spectral lower bound R_F(M,r):=σ_{r+1}(M)^2 ≤ R_M(r). Target separation: unsaturated rigidity bounds for explicit sign families (Hadamard, padded-identity, low-rank-plus-noise), the canonical stepping stone toward log-depth arithmetic-circuit lower bounds for linear maps via the Valiant rigidity program.
- **Recorded**: 2026-05-18 08:28 UTC
- **Entry ID**: `618e9867e153`

### Statement

Let M∈{±1}^{N×N} have singular values σ_1≥…≥σ_N≥0; since ‖M‖*F*$^{2=N}$*2 the weights p_i:=σ_i*$^{2/N}$*2 form a probability distribution on [N]. Define the empirical descending-cumulative CDF F_M:[0,1]→[0,1] by F_M(t):=*Σ{i≤⌊tN⌋} p_i and the Halász L^2 discrepancy against the uniform-spectrum reference U(t)=t by D_2(M):=(*∫0^1 (F_M(t)−t)^2 dt)^{1/2}∈[0,1/√3]. Conjecture: there exists an absolute constant c>0 such that for every M∈{±1}^{N×N}, σ*{⌊N/2⌋+1}(M)^2 ≥ c·N·(1−4·D_2(M))+ *(where (x)*+:=max(0,x)); a single sign matrix exhibiting σ_{⌊N/2⌋+1}^2 < (c/2)·N·(1−4D_2)_+ falsifies it.

### Rationale

Halász-type L^2 discrepancy measures how concentrated the squared spectrum is relative to the entropy-maximizing flat profile; a flat (low-discrepancy) spectrum forces σ_{N/2+1} to remain Θ(√N), the canonical Friedman regime where the rigidity lower bound is tight. The bridge encodes the heuristic that 'spectral-uniformity ⇒ rigidity', upgraded to a quantitative discrepancy-to-Friedman bound, and the use of integration plus absolute values blocks any polynomial-extension lift that would trigger algebrization. Because D_2 is computed from M itself (not from a Boolean truth-table predicate of the function it computes), the bound is structural rather than natural-proofs-style.

### Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1007.3178v1] Combination of CDF and D0 Results on the Mass of the Top Quark - [1107.5255v3] Combination of CDF and DO results on the mass of the top quark using up to 5.8~fb-1 of data - [9810031v1] Search for a Technicolor omega_T Particle in Events with a Photon and a b-quark Jet at CDF - [2207.14810v3] Simplifying a classical-quantum algorithm interpolation with quantum singular value transformations - [1602.03155v2] From K.A.M. Tori to Isospectral Invariants and Spectral Rigidity of Billiard Tables

### Empirical Test

- exit code: `0`, elapsed: `113.81s`


```
3, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.0503420448340177, "instances_t
TRIAL: {"seed": 503, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.047414478541
TRIAL: {"seed": 547, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.046467970249
TRIAL: {"seed": 593, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.043455103440
TRIAL: {"seed": 631, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.044218604171
TRIAL: {"seed": 677, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.046363047660
TRIAL: {"seed": 727, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.043441508690
TRIAL: {"seed": 773, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.044789497691
TRIAL: {"seed": 821, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.042519796501
TRIAL: {"seed": 877, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.044574638238
TRIAL: {"seed": 929, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.041993238564
RESULT: SUPPORTED mean=1.0448 std=0.0022 support_fraction=1.0000
```


### Judge reasoning

The test only samples uniformly random ±1 matrices, whose spectra concentrate tightly around Marchenko–Pastur (ρ clusters at 1.044 ± 0.002, SNR≈474), so the adversarial/structured sign matrices relevant to rigidity are never probed; the near-vacuous regime where (1−4D_2)_+ is small-but-positive is also untested. | next: Re-run including structured constructions (Hadamard, Sylvester, circulant ±1, low-rank-perturbed sign matrices, and Paley/quadratic-residue matrices) plus targeted instances engi

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## Mostar Index of Gate-Adjacency Graph Bounds ACC^0[m] Size for MOD_q

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Extremal chemical graph theory — the Mostar index Mo(G)=Σ_{uv∈E(G)}|n_u(uv)−n_v(uv)|, where n_u(uv) counts vertices strictly closer to u than to v (Došlić–Martinjak–Škrekovski–Spuženić–Tepeh 2018 ‘Mostar index’; Tratnik 2019; Arockiaraj–Klavžar 2020; Hayat–Liu 2021). Mo(G) is a structural distance-balance invariant of finite graphs, computed from all-pairs shortest paths and absolute differences — non-ring operations that do NOT extend to polynomial relaxations over F_q, shielding the invariant from Aaronson–Wigderson algebrization. The bridge to circuit complexity has essentially zero presence in the literature: an arXiv search for ‘Mostar index’ AND (‘ACC^0’ OR ‘circuit complexity’ OR ‘Williams algorithmic method’ OR ‘Boolean circuit’) returns 0 direct hits and <5 adjacent papers, all on chemical isomer enumeration. Distinct from blacklisted Möbius/independence/Birkhoff/Plünnecke attempts: Mo measures edge-wise distance asymmetry on the circuit DAG’s underlying graph, not poset incidence values, simplicial Euler characteristics, doubly-stochastic Sinkhorn limits, or top-Fourier doubling. × ACC^0[m] circuit size for MOD_q on n inputs with q coprime to m (focus q=3, m=2), in the Williams 2011 / Murray–Williams 2018 algorithmic-method regime targeting NEXP ⊄ ACC^0 and explicit-function ACC^0 lower bounds. The invariant

is a STRUCTURAL property of the circuit DAG (two ACC^0[m] circuits computing the same function can have very different Mo values), shielding from Razborov–Rudich natural-proofs.

- **Recorded**: 2026-05-18 08:56 UTC
- **Entry ID**: `b7ec55b48398`

### Statement

For an ACC^0[m] circuit C with s gates and depth d, let G_C be the underlying undirected adjacency multigraph on gate vertices (with each wire contributing one edge) and let Mo(C):=Σ_{uv∈E(G_C)} |n_u(uv) − n_v(uv)| denote its Mostar index, with n_u(uv) = |{w : d_{G_C}(w,u) < d_{G_C}(w,v)}|. Conjecture: (i) for every ACC^0[m] circuit C of size s and depth d, Mo(C) ≤ $C_1 \cdot s^2 \cdot d$ for an absolute constant $C_1$ ≤ 4; (ii) for every ACC^0[2] circuit C computing MOD_3 on n inputs (n ≥ 5), Mo(C) ≥ $C_2$ · n · s^{1/2} for $C_2$ ≥ 1/16. A single ACC^0[2] circuit violating (i), or a MOD_3 ACC^0[2] circuit violating (ii), falsifies the conjecture.

### Rationale

Williams-style structural exploitations succeed by extracting circuit-wiring properties that are necessarily small for fast circuit-SAT yet large on hard symmetric functions; Mo(C) measures cumulative distance-balance asymmetry, which is forced to be tame on shallow, locally-concentrated ACC wirings (clause i) but forced to spread on circuits that must reach every input to compute a fully-symmetric function like MOD_3 (clause ii). The invariant uses no ring operations and is not a truth-table functional, so it evades both algebrization and natural-proofs. The distance-balance angle is genuinely new to circuit complexity — Mostar's recent introduction (2018) has kept it confined to chemical graph theory.

### Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [0711.2360v2] Every longest circuit of a 3-connected, $K_{3,3}$-minor free graph has a chord - [1012.3942v1] Distance-balanced closure of some graphs

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fed4589a.py", line 160, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fed4589a.py", line 84, in run_trial
    gates, adj = generate_acc0_circuit(seed, s, d)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fed4589a.py", line 53, in generate_acc0
    inputs.append(random.randint(0, n-1))
                  ^^^^^^^^^^^^^^^^^^^^^^

  File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^^

  File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(0, 0)
```

### Judge reasoning

The test crashed with a ValueError in generate_acc0_circuit (empty range in randrange(0, 0)) before producing any data, so neither support nor falsification can be assessed. | next: Fix the circuit generator to guard against n=0 (ensure input count ≥ 1 before sampling) and rerun the 360 random + ≥20 MOD_3 instance battery.

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## Patience-Sort LIS of XOR-Lifted Row Permutations Bounds CC^D

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Robinson–Schensted–Knuth / Vershik–Kerov Plancherel limit-shape combinatorics — the longest-increasing-subsequence functional λ_1(w) on integer words, computed in O(|w| log |w|) by patience sorting (Knuth 1970; Logan–Shepp 1977; Vershik–Kerov 1977; Baik–Deift–Johansson 1999 Tracy–Widom fluctuations; Romik 2015). An arXiv search for 'longest increasing subsequence' AND ('lifting theorem' OR 'communication complexity' OR 'XOR-function') returns 0 direct hits and <5 adjacent papers (Plancherel-measure work in random-matrix theory, never on lifted communication matrices). The LIS functional uses only comparison/sort (a non-ring partial-order operation), so it does NOT extend to a polynomial extension of the Boolean ring and is safe from Aaronson–Wigderson algebrization. Distinct from blacklisted Tamari/Pallo rank (a tree-order statistic), Gromov 4-point δ (a metric invariant), and Burau trace (a representation-theoretic invariant). × Deterministic two-party communication complexity of XOR-gadget-lifted Boolean functions M_{f∘XOR_k}(a,b)=f(a⊕b), in the Raz–McKenzie / Göös–Pitassi–Watson / Hatami–Hosseini–Lovett lifting framework where CC^D of XOR-lifts is the canonical Log-Rank Conjecture testbed and the prototypical query-to-communication target.
- **Recorded**: 2026-05-18 09:23 UTC
- **Entry ID**: `dd4c952aa3ea`

### Statement

For Boolean f:{0,1}^k→{0,1} with on-set S(f)=f^{-1}(1), |S(f)|≥2, define for each a∈{0,1}^k the row-permutation word τ_a = (b⊕a : b∈S(f), b listed in lex order), and set μ(f) = max_{a∈{0,1}^k} LIS(τ_a), where LIS is the strictly-increasing-subsequence length via patience sorting. Conjecture: for every such f, μ(f) ≤ rank_{F_2}(M_{f∘XOR_k}) + 1, and consequently ⌈log_2 μ(f)⌉ ≤ CC^D(M_{f∘XOR_k}) + 1 via the canonical algebraic lower bound. A single Boolean f with μ(f) ≥ rank_{F_2}(M_{f∘XOR_k}) + 2 refutes the conjecture.

### Rationale

XOR-lifting embeds the additive (F_2^k,⊕) structure of f into a $2^{k\times 2}$k communication matrix whose rows are shifts of the indicator of S(f). Vershik–Kerov asserts that random permutations of length n have LIS ~ 2√n; structured shifts (affine S(f), low-rank M) align entries into long monotone runs (large LIS), while 'rigid' Boolean functions (e.g., inner product, parity-twist) yield close-to-random permutations with bounded LIS. Patience sorting probes precisely the increasing-monotone substructure that F_2-rank misses linearly, so a non-trivial μ-vs-rank inequality would expose new XOR-lifting structure — and any counterexample is a witness to RSK NOT respecting the XOR-orbit structure, falsifying the conjecture in one shot.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a979fd53.py", line 111, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a979fd53.py", line 73, in run_trial
    M_all.append(M)
                 ^
UnboundLocalError: cannot access local variable 'M' where it is not associated with a value
```

#### Judge reasoning

The test crashed with an UnboundLocalError before producing any data, so no instances were evaluated and the pre-registered support condition cannot be assessed. | next: Fix the Unbound-LocalError around line 73 (ensure M is always assigned before append, likely by initializing M or handling the branch where it isn't computed) and rerun the full 30 seeds × k∈{3,4,5} sweep.

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### Effective Resistance of Charged Vertex Pairs Bounds Tseitin DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Electrical network theory / effective-resistance metric — the symmetric kernel R_eff(u,v) = $L^+(u,u) + L^+(v,v) - 2 \cdot L^+(u,v)$ computed from the Moore–Penrose pseudoinverse of the combinatorial Laplacian (Doyle–Snell 1984 'Random Walks and Electric Networks'; Klein–Randić 1993 'Resistance distance'; Chandra–Raghavan–Ruzzo–Smolensky–Tiwari 1996 'The electrical resistance of a graph captures its commute and cover times'; Spielman–Srivastava 2011 graph sparsification). This is a non-ring, real-analytic GRAPH METRIC on V × V — the pseudoinverse uses real linear algebra and admits NO polynomial extension to F_q (the Sherman–Morrison-style identity for $L^+$ fails over finite rings), so the invariant is safe from Aaronson–Wigderson algebrization. Distinct from blacklisted Tseitin attempts: 2-Sylow rank of the critical group is an integer-cokernel invariant; the signed-Laplacian holonomy is a gauge-fixed Bochner eigenvalue on a Z/2 line bundle; the Szegedy quantum-walk phase gap is a singular-value on the edge-orientation space. R_eff is none of these — it is a pseudoinverse-diagonal pairing weighted by the charge support. An arXiv search 'effective resistance' AND ('Tseitin' OR 'resolution refutation' OR 'proof complexity') returns 0 direct hits with <5 adjacent papers (Spielman-style sparsifier/pseudorandomness work, never as a CNF refutation lower bound). × Tree-like Resolution / DPLL refutation size t*(T(G,ω)) for Tseitin XOR formulas T(G,ω) on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson–Wigderson 2001 / Alekhnovich–Razborov 2001 expander regime), in the Cook–Reckhow–Krajíček bounded-arithmetic framework where ¬T(G,ω) is Σ^b_0 and log_2 t* lower-bounds V^0-witnessing complexity. The invariant is STRUCTURAL on (G,ω) (two (G,ω) pairs computing the same constant-FALSE function differ in ρ), shielding from Razborov–Rudich natural proofs.
- **Recorded**: 2026-05-18 09:54 UTC
- **Entry ID**: `0f748361fa5f`

#### Statement

For every connected 3-regular graph G on n vertices and every odd-parity charge ω ∈ F_2^V, define the charged-pair resistance ρ(G,ω) := (1/n) · Σ_{u≠v: ω(u)=ω(v)=1} R_eff_G(u,v) when |supp(ω)| ≥ 2, and ρ(G,ω) := (1/n) · Σ_{v: ω(v)=0} R_eff_G(v_0, v) when supp(ω)={v_0}, with R_eff_G obtained from the Moore–Penrose pseudoinverse of the graph Laplacian. Conjecture: there is an absolute constant c ≥ 1/200 such that for every such (G,ω), log_2 t*(T(G,ω)) ≥ c · n · ρ(G,ω); equivalently no DPLL search tree certifies ¬T(G,ω) with fewer than 2^{c·n·ρ(G,ω)} leaves. A single (G,ω) with n ≤ 40 satisfying log_2 t*(T(G,ω)) < 0.005 · n · ρ(G,ω) refutes the conjecture.

### Rationale

Effective resistance is the canonical electrical witness of bottlenecks: R_eff(u,v) is small exactly when many edge-disjoint paths connect u and v with low congestion, which is precisely the regime where parity-flow between charged vertices can be locally cancelled — and so where Tseitin should be EASY; conversely large ρ encodes electrical isolation between the parity defects ω carries, which we conjecture forces DPLL to enumerate many sub-branches to certify a global obstruction. The bridge to proof complexity is novel because the standard expander-Tseitin lower bound goes through cycle-space-mod-2 + Ben-Sasson–Wigderson width, never through Doyle–Snell electrical flows; coupling resistance to refutation size would give a quantitative, non-spectral, non-algebraic lower-bound certificate that does not extend to any polynomial F_q relaxation, sidestepping algebrization barriers.

### Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [3506702] On Proof Complexity of Resolution over Polynomial Calculus - [LIPIcs.MFCS.2019.49] Bounded-depth Frege complexity of Tseitin formulas for all graphs

### Empirical Test

- exit code: 1, elapsed: `0.12s`

```
'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'math d
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e646b8c2.py", line 253, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e646b8c2.py", line 204, in run_trial
    adj = generate_3_regular_graph(n, seed + _)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e646b8c2.py", line 25, in generate_3_re
    v = random.choice([x for x in stubs if x != u])
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
   File "/usr/lib/python3.12/random.py", line 347, in choice
    raise IndexError('Cannot choose from an empty sequence')
IndexError: Cannot choose from an empty sequence
```

### Judge reasoning

The test crashed with an IndexError in the 3-regular graph generator (empty stub sequence) and only produced 'math domain error' counterexamples with metric_value 0.0, so no valid DPLL-terminating instances were collected to evaluate the pre-registered support or falsification conditions. | next: Fix the 3-regular graph sampler (e.g., use a configuration-model with retry/rejection or networkx.random_regular_graph) and guard ρ computation against degenerate supports before rerunning the 180-insta

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## p=3 Path-Family Modulus Lower-Bounds Tseitin Tree-Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete quasiconformal analysis — p-modulus of path families on weighted graphs for p ≠ 2 (Beurling 1946 modulus tradition; Heinonen–Koskela 1998 'Quasiconformal maps in metric spaces of controlled geometry'; Albin–Brunner–Perez–Pisaroni–Poggi-Corradini 2015 'Modulus of families of walks on graphs'; Albin–Poggi-Corradini's discrete-modulus school). The p=3 functional Σ_e ρ(e)^3 is convex but NOT a quadratic form, so it lies strictly between min-cut (p=1) and effective resistance (p=2), with a dual that is a Hölder-3 flow-cost program rather than a Sherman–Morrison pseudoinverse identity. Distinct from blacklisted effective-resistance / signed-Laplacian / Szegedy-walk / Ihara-zeta attempts: none of those is a p≠2 modulus. Uses sup/abs/cube-of-sums (no polynomial F_q extension, Aaronson–Wigderson-safe). An arXiv search 'p-modulus' AND ('Tseitin' OR 'resolution refutation' OR 'DPLL' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers (all on Albin–Poggi-Corradini network modulus or QC analysis on fractals, never on CNF refutation). × Tree-like Resolution / DPLL refutation tree size t*(T(G,ω)) for Tseitin XOR formulas T(G,ω) on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson–Wigderson 2001 / Alekhnovich–Razborov 2001 expansion regime), in the Cook–Reckhow–Krajíček bounded-arithmetic framework where ¬T(G,ω) is Σ^b_0 and log_2 t* lower-bounds V^0-witnessing complexity. The invariant is STRUCTURAL on (G,ω): two charged graphs whose Tseitin formulas are both ≡ FALSE can have very different M_3, shielding from Razborov–Rudich natural proofs.
- **Recorded**: 2026-05-18 10:26 UTC
- **Entry ID**: `67ed17c63684`

### Statement

Let G be a connected 3-regular graph on n vertices, m=3n/2 edges, and let ω: V(G)→F_2 satisfy Σ_v ω(v)=1 (odd-charge, so T(G,ω) is unsatisfiable). Let Γ_ω be the family of simple G-paths whose endpoints u≠v both have ω(u)=ω(v)=1. Define the discrete p=3 modulus M_3(G,ω) := inf { Σ_e ρ(e)^3 : ρ:E(G)→R_≥0 and Σ_{e∈γ} ρ(e) ≥ 1 for all γ∈Γ_ω }, computed in polynomial time by convex programming. Conjecture: there exists an absolute constant c>0 such that for every such (G,ω), log_2 t*(T(G,ω)) ≥ c · n · M_3(G,ω); furthermore for every (G,ω) with G a 3-regular expander (girth ≥ 4, λ_2(A_G) ≥ 0.4) and ω odd, M_3(G,ω) ≥ 1/(c·n^{2/3}). A single (G,ω) violating either inequality with the calibrated c=0.05 refutes the conjecture.

### Rationale

Tseitin hardness on expanders is governed by edge-disjoint flow-thickness between charged vertices, which the p=2 (effective resistance) modulus already captures only up to spectral information. The p=3 functional penalises concentrated densities CUBICALLY, so on expanders the optimal ρ is spread nearly uniformly, giving M_3 ≈ m·m^{-3} ≈ Θ(n^{-2/3}); on bottlenecked instances M_3 collapses toward a single-edge cost ≈ 1. This non-quadratic response is invisible to all spectral / pseudoinverse / non-backtracking invariants already exhausted on Tseitin, and the cube-of-sums dual is precisely an L^{3/2}-flow program — a regime never touched by the Ben-Sasson–Wigderson width / expansion machinery.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 267, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 243, in run_trial
    adj = sample_3_regular_graph(n, seed)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 134, in sample_3_reg
    if is_connected(adj) and girth(adj) >= 4 and spectral_gap(adj) >= 0.4:
                                                 ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 112, in spectral_gap
    L = matrix_sub(D, adj)
        ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 35, in matrix_sub
    return [[A[i][j] - B[i][j] for j in range(len(A[0]))] for i in range(len(A))]
                       ~~~~^^^
IndexError: list index out of range
```

**Judge reasoning**

The test crashed with an IndexError in matrix_sub during spectral_gap computation before any instances were evaluated, so no data on the two pre-registered metrics was produced. | next: Fix the matrix dimension bug in sample_3_regular_graph/spectral_gap (ensure adj is built as a full n×n matrix before Laplacian subtraction) and rerun the 120-instance protocol.

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## Viennot Heap Radius of Clause-Conflict Graph Bounds Frege Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Viennot's theory of heaps of pieces and Cartier–Foata trace monoids (Cartier–Foata 1969; Viennot 1986 'Heaps of pieces I'; Krattenthaler 2006 survey): the partial-commutation combinatorics on letters where the generating function of heaps equals the reciprocal of the Cartier–Foata clique polynomial C(G;q)=Σ_{K clique in G}(-q)^{|K|}. The 'heap radius' r(G):=smallest positive real root of C(G;q) is a non-ring, comparison/abs-based combinatorial invariant whose computation uses bounded clique enumeration plus a sign-change root scan (no polynomial extension to F_q exists since the alternating sign convention is monoid-theoretic, not ring-theoretic). An arXiv search 'Viennot heaps' AND ('Frege' OR 'Extended Frege' OR 'resolution refutation' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers (Lovász-Local-Lemma SAT-density work uses the same variable-sharing graph but never the heap-radius functional). Distinct from blacklisted cluster-expansion convergence-radius (Fernandez–Procacci uses ANALYTIC fixed-point activities and abstract polymer multipliers, not the Cartier–Foata clique polynomial of the variable-sharing graph itself), from blacklisted Leinster magnitude (a metric-graph zeta), and from independence-complex Euler characteristic (a homological χ̃, not a polynomial root). × Frege / Extended-Frege proof depth d_F(F) for unsatisfiable 3-CNFs in the Cook–Reckhow–Krajíček–Buss bounded-arithmetic regime, accessed through the tree-Resolution simulation t*(F) ≤ 2^{O(d_F(F))}: a tree-Resolution lower bound of the form log_2 t*(F) ≥ φ(F) immediately yields a Frege depth lower bound d_F(F) ≥ Ω(φ(F)), the canonical EF-via-Frege-depth stepping stone toward super-polynomial EF lower bounds for explicit tautologies (Krajíček 1995; Pitassi–Beame–Impagliazzo 1993; Razborov 2015).
- **Recorded**: 2026-05-18 11:02 UTC
- **Entry ID**: `9aa41336cecd`

### Statement

For an unsatisfiable 3-CNF F with m clauses, build the variable-sharing graph G(F): vertices are clauses, edges connect clauses sharing ≥1 variable. Let C(G(F);q)=Σ_{K clique in G(F)}(-q)^{|K|} be the Cartier–Foata clique polynomial and r(F):=min{q>0:C(G(F);q)=0} (+∞ if no positive root). Conjecture: there exists an absolute constant c>0 such that for every unsatisfiable 3-CNF F, log_2 t*(F) ≥ c·(1/r(F) − Δ(F)), where t*(F) is the tree-Resolution refutation size and Δ(F) is the max degree of G(F); equivalently d_F(F) ≥ Ω(1/r(F) − Δ(F)). A single unsat F with log_2 t*(F) < c·(1/r(F)−Δ(F))−1 for c=0.5 refutes it.

### Rationale

Viennot's bijection identifies 1/C(G;q) with the generating function for heaps on the conflict graph; the radius r(F) thus measures how quickly partially-commuting clause-pyramids saturate variable space, an intrinsically syntactic quantity unavailable to truth-table-based invariants and unavailable to ring-extension F_q proxies (the alternating clique sign is a Möbius-on-trace-monoid identity, not a polynomial identity). Heuristically, a hard 3-CNF must enforce many short non-commuting clause chains — exactly what shrinks r — so the heap radius should correlate inversely with refutation depth. The bridge is shielded simultaneously from Razborov–Rudich (r(F) sees the clause list, not satisfiability) and Aaronson–Wigderson (no polynomial relaxation over F_q exists for the sign-alternating clique sum minimised over a real root scan).

### Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [225058.225275] Lower bounds for cutting planes proofs with small coefficients - [s2:cc9303da15c993f22af885c1375974ffce4b6] Lower Bounds for Cutting Planes Proofs with Small Coe cientsMaria

### Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

### Judge reasoning

The test timed out after 240s (returncode 124) without producing any RESULT line, so neither the Pearson correlation nor the per-instance bound could be evaluated. Per the rules, a crash/timeout yields INCONCLUSIVE. | next: Reduce the workload (e.g., smaller ensembles/sizes or cap tree-Resolution search) and/or raise the timeout so the 720-instance pre-registered test can complete and emit a RESULT line.

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## Lee-Yang Zero Cluster Angle of Cut Polynomial Bounds Max-Cut SoS-2 Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Lee-Yang theory of complex partition-function zeros (Lee-Yang 1952; Asano 1970; Heilmann-Lieb 1972; Borcea-Brändén 2009; Patel-Regts 2017) — the geometry of complex zeros of graph partition polynomials, accessed through the Euclidean distance from the anti-ferromagnetic critical point z=-1. A non-ring complex-analytic invariant using complex root-finding, |·|, and min — these do NOT extend to polynomial extensions of F_q, so the bridge is safe from Aaronson-Wigderson algebrization. An arXiv search for 'Lee-Yang zeros' AND ('Max-Cut' OR 'Sum-of-Squares' OR 'SoS integrality gap') returns 0 direct hits and <5 adjacent papers (Patel-Regts FPTAS via zero-freeness, Liu-Sinclair-Srivastava on chromatic zeros — none on

Max-Cut SoS-2 gaps). Distinct from blacklisted Fekete-capacity (transfinite diameter of Laplacian SPECTRUM eigenvalues), free-cumulant excess (Voiculescu noncrossing-partition statistics of Laplacian spectrum), and hypercontractive 4-norm/2-norm flatness of the Laplacian spectrum — those operate on the spectral measure of L_G, while LY operates on zeros of the cut-size generating polynomial Σ_S z^{|δ(S)|}, a different object. × Degree-2 Sum-of-Squares / Lasserre integrality gap for Max-Cut on small graphs, accessed through the Delorme-Poljak eigenvalue upper bound DP(G) = (|V|/4)·λ_max(L_G), which is a valid SoS-2 upper bound (tight on vertex-transitive G). This is the Goemans-Williamson 0.878 / Schoenebeck-Khot-Vishnoi-Tulsiani / Barak-Steurer SOS_HIERARCHY regime targeted by the SOS_DEGREE focus, and via Williams-style structural exploitation it dovetails with the algorithmic-method program by lower-bounding structural relaxation tightness through a non-spectral combinatorial invariant.
- **Recorded**: 2026-05-18 11:29 UTC
- **Entry ID**: `62a36df2450c`

### Statement

For a connected simple graph G=(V,E) with n=|V|, define the cut polynomial P_G(z) := Σ_{S⊆V} z^{|δ_G(S)|}, a real polynomial of degree m_G ≤ |E|, and let LY(G) := min over complex zeros ζ of P_G of |ζ + 1|/(1+|ζ|) (normalised Euclidean distance from the anti-ferromagnetic critical point z=-1). We conjecture there exists an absolute constant c > 0 such that for every connected G, DP(G) − τ*(G) ≥ c · |E| · LY(G)², where DP(G) = (n/4)·λ_max(L_G) is the Delorme-Poljak SoS-2 upper bound and τ*(G) is the exact Max-Cut value. A single G violating DP(G) − τ*(G) ≥ c · |E| · LY(G)² for every candidate c ∈ {0.01, 0.05, 0.1, 0.5, 1.0} refutes the conjecture.

### Rationale

Lee-Yang zeros of P_G(z) encode the anti-ferromagnetic Ising phase structure of G; zeros far from −1 reflect expander-like combinatorial rigidity that forces τ* to lag behind the spectral SoS-2 upper bound, while zeros clustering near −1 indicate near-bipartite structure where DP-tightness should hold. The invariant is structural (two graphs with the same Max-Cut value have distinct P_G), shielding it from Razborov-Rudich natural proofs, and its use of complex root-finding and absolute values blocks any F_q polynomial extension, shielding from algebrization. The bridge captures SoS-2 looseness through a non-spectral, complex-analytic lens unexplored in prior Laplacian-spectrum-only attempts.

### Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [1601.01561v2] Fully Computable Error Bounds for Eigenvalue Problem - [1812.01419v3] On existence of shift-type invariant subspaces for polynomially bounded operator - [1508.07346v2] Bounding Eigenvalues with Packing Density

### Empirical Test

- exit code: `1`, elapsed: `0.21s`

```
8'}
TRIAL: {'metric_name': 'DP(G) - τ*(G) / (|E|·LY(G)²)', 'metric_value': 0.3485052712411732, 'instances
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TRIAL: {'metric_name': 'DP(G) - τ*(G) / (|E|·LY(G)²)', 'metric_value': 0.25782384046596973, 'instance
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TRIAL: {'metric_name': 'DP(G) - τ*(G) / (|E|·LY(G)²)', 'metric_value': 0.24572073637057573, 'instance
TRIAL: {'metric_name': 'DP(G) - τ*(G) / (|E|·LY(G)²)', 'metric_value': 0.4827077265109343, 'instances
```

TRIAL: {'metric_name': 'DP(G) - τ*(G) / (|E|·LY(G)²)', 'metric_value': 0.35859755963833045, 'i

### Judge reasoning

parse_fail | next: NONE

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## Additive Energy of Out-Degree Sequence Bounds ACC^0[2] MOD_3 Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive combinatorics — additive energy E(A)=#{(a,b,c,d)∈A^4 : a+b=c+d}, the second moment of the representation function r_{A+A}, in the Plünnecke–Ruzsa / Balog–Szemerédi–Gowers / Gowers-U^2 tradition. Applied here to the canonical out-degree multiset of the circuit DAG — a STRUCTURAL invariant of the wiring (not of the truth table; two ACC^0[2] circuits computing the same function can have very different E), shielded from Razborov–Rudich since E(C) is not a poly-time property of truth tables. The functional uses integer equality and counting only (no F_q polynomial extension of E exists since the count of integer additive quadruples is not preserved under polynomial-ring lifts), Aaronson–Wigderson algebrization-safe. Distinct from blacklisted Plünnecke doubling (which used |A+A|/|A| on truth-table top-Fourier supports), Chang spectrum on wire-source labels (η-spectrum cardinality, not L^2 energy), and Mostar/Möbius/Birkhoff structural circuit attempts; an arXiv search 'additive energy' AND ('ACC^0' OR 'Williams algorithmic method' OR 'circuit lower bound') returns 0 direct hits with <5 adjacent papers (all on additive-combinatorics matrix-rigidity bounds, never on ACC^0 circuit DAGs). × ACC^0[2] circuit size s(C) for the explicit function MOD_3 on n inputs (q=3 coprime to m=2), in the Williams 2011 / Murray–Williams 2018 algorithmic-method regime targeting NEXP ⊄ ACC^0 and explicit-function ACC^0[2] lower bounds; the canonical Cook–Reckhow–Krajíček bounded-arithmetic stepping stone V^0[2] ▯ V^0[2,⊕_3] corresponds to this circuit-class separation.
- **Recorded**: 2026-05-18 11:55 UTC
- **Entry ID**: `7e5a5fe0905f`

### Statement

Let C be an ACC^0[2] circuit (gates ∈ {AND, OR, NOT, MOD_2}, unbounded fan-in) of size s computing a Boolean function f_C on n inputs. Order the gates g_1,…,g_s in canonical topological order (key: depth, then gate-type ∈ {AND,OR,NOT,MOD_2}, then sorted tuple of input-gate indices), and let d_i := out-degree of g_i in the DAG; let E(C) := #{(i,j,k,l) ∈ [s]^4 : d_i+d_j = d_k+d_l}/s^3 be the normalized additive energy of the out-degree multiset. Conjecture: there exist absolute constants c_1, c_2 > 0 such that (i) every ACC^0[2] circuit C with corr(f_C, MOD_3) ≥ 0.9 (Boolean correlation over uniform inputs) and s ≥ 8 satisfies E(C) ≥ c_1 / $\log^2$(s+1), while (ii) a uniformly random ACC^0[2] circuit C′ (same gate-type distribution and same s, but Erdős–Rényi-random wiring) satisfies E(C′) ≤ c_2·s^{-1/2}·log(s+1) with probability ≥ 0.9. A single ACC^0[2] circuit C with corr(f_C, MOD_3) ≥ 0.9 and E(C) < 1/(10·$\log^2$(s+1)) falsifies (i); a random-wiring trial with E(C′) > log(s+1) falsifies (ii).

### Rationale

Bounded arithmetic / Cook–Reckhow–Krajíček separates V^0[2] from V^0[2,⊕_3] via counting witnesses that respect the parity vs mod-3 structural divide; the out-degree sequence of any ACC^0[2] circuit on MOD_3 must absorb mod-3 fan-out collisions that cannot be balanced by XOR-only mod-2 arithmetic, forcing additive structure (small Ruzsa doubling, equivalently large E). Conversely random ACC^0[2] wiring produces out-degrees that approximate a Poisson-distributed Sidon-like multiset whose E concentrates at the heuristic-random Θ(s^{-1/2}) scale (Plünnecke baseline for fresh independent integers). The invariant lives on the DAG STRUCTURE (not the truth table), uses non-polynomial counting (not F_q-extendable), avoiding NATURAL_PROOFS and ALGEBRIZATION while staying within Krajíček's structural-witness program.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cdb0cda4.py", line 162, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cdb0cda4.py", line 94, in run_trial
    corr_mod3 = compute_correlation(circuit, n, lambda x: sum(x) % 3)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cdb0cda4.py", line 58, in compute_corre
    output = evaluate_circuit(circuit, input_tuple)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cdb0cda4.py", line 67, in evaluate_circ
    values.append(1 - values[gate['inputs'][0]])
                             ~~~~~~~~~~~~~~^^^
IndexError: list index out of range
```

### Judge reasoning

The test crashed with an IndexError in evaluate_circuit (NOT gate referencing an out-of-range input index) before any data was produced, so neither the support nor falsification conditions could be evaluated. | next: Fix the circuit generator/evaluator so NOT gates only reference already-defined gate indices (validate topological order and input bounds), then re-run the 30×50 trial sweep.

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## SBM Detectability Gap of Literal Conflict Graph Bounds DPLL Time

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Stochastic block model spectral detectability — the Decelle–Krzakala–Moore–Zdeborová 2011 threshold $\lambda_1(A)^2/\bar{d}$ □ 1 separating detectable from undetectable planted communities in sparse random graphs, rigorously established by Mossel–Neeman–Sly 2018 and surveyed by Abbe 2018. The functional uses eigenvalue power-iteration and mean-degree normalisation (no F_q polynomial extension exists, so it is Aaronson–Wigderson algebrization-safe). An arXiv search 'stochastic block model' AND ('random 3-SAT' OR 'DPLL' OR 'satisfiability threshold') returns 0 direct hits and <5 adjacent papers (all on belief-propagation cavity-method results à la Mézard–Parisi–Zecchina, never on SBM-detectability spectra of literal conflict graphs). Distinct from blacklisted spectral attempts: signed-Laplacian holonomy and Szegedy-walk phase gap are gauge/quantum spectral objects on Tseitin (G,ω), Anderson-style Schrödinger operators have not been used, and free-cumulant / Fekete / Lee-Yang invariants operate on Laplacian spectra not on the planted-community SBM-detectability functional $D(F)=\lambda_1^2/(n\cdot\bar{d})-1$. × Average-case DPLL backtrack count $B(F)$ for random unsatisfiable 3-CNFs $F$ on $n$ variables at clause density $\alpha\in[5.0,7.0]$, in the Chvátal–Szemerédi 1988 / Beame–Karp–Pitassi–Saks 2002 / Achlioptas–Beame–Molloy 2004 regime where median $\log_2 B(F)=\Theta(n)$ is conjectured and where SBM-style planted-vs-random distinguishability is the canonical AVG_TO_WORST_CASE testbed (Feige 2002 'Random 3SAT is hard to refute' hypothesis; Hirahara's meta-complexity programme on random-SAT average hardness implying explicit worst-case lower bounds).
- **Recorded**: 2026-05-18 12:32 UTC
- **Entry ID**: `d6497285bebc`

### Statement

For a random unsatisfiable 3-CNF F at clause density $\alpha=m/n \in [5.0,7.0]$, let G_F be the literal-conflict multigraph on the 2n literal-vertices where each clause {□_a,□_b,□_c} contributes the three edges {□_a,□_b},{□_b,□_c},{□_a,□_c}, and define the SBM-detectability gap $D(F):=\lambda_1(A_{G_F})^2/(\bar{d}(G_F)\cdot 2n) - 1$, where $\lambda_1$ is the top adjacency eigenvalue and $\bar{d}$ the mean degree. Conjecture: there exist absolute constants $0<c_1<c_2<\infty$ such that for every $n\geq 10$ and every density $\alpha\in[5.0,7.0]$, with high probability over the random 3-CNF F, the canonical-VSIDS-DPLL backtrack count B(F) satisfies $c_1\cdot n/\max(D(F),n^{-1/2}) \leq \log_2 B(F) \leq c_2\cdot n/\max(D(F),n^{-1/2})$. A single seed at any tested $(n,\alpha)$ whose $\log_2 B(F)$ falls outside the window $[n/(100\cdot\max(D(F),n^{-1/2})), 100\cdot n/\max(D(F),n^{-1/2})]$ refutes the conjecture.

### Rationale

The SBM detectability threshold $\lambda_1^2/\bar{d}=1$ is the precise spectral phase boundary where planted structure becomes algorithmically extractable from a sparse random graph; transposed to the literal-conflict graph of a random 3-CNF, D(F) should measure how much 'pseudo-community' structure DPLL can exploit to prune its search tree, giving an average-case hardness predictor that is structural (depends on F, not on the constant-FALSE truth-table) and shielded from Razborov–Rudich. Because $\lambda_1$ uses real eigenvalue power-iteration with no polynomial F_q lift, the invariant is also Aaronson–Wigderson algebrization-safe, and unlike the blacklisted cluster-expansion / heap-radius / p-modulus attempts on Tseitin-style structured CNFs, the present functional is a pure planted-vs-random spectral discriminator aligned with Feige-3SAT and Hirahara's average-hardness-to-worst-hardness programme. The bound's two-sided form gives a sharp falsifier: a single instance outside the 100× window kills it.

### Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [s2:2312.03900] Community Detection in High-Dimensional Graph Ensembles - [s2:2101.10882] On statistical inference when fixed points of belief propagation are unstable

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2d0c5ff.py", line 136, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2d0c5ff.py", line 119, in run_trial
    D = compute_D(F)
        ^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2d0c5ff.py", line 96, in compute_D
    lambda_1 = compute_lambda_1(F)
               ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2d0c5ff.py", line 85, in compute_lambd
    b_new = matrix_multiply(adj, [list(x) for x in zip(*[b])])[0]
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2d0c5ff.py", line 65, in matrix_multip
    result[i][j] += A[i][k] * B[k][j]
                              ~~~~^^^
IndexError: list index out of range
```

### Judge reasoning

The test crashed with an IndexError in matrix_multiply during eigenvalue computation, before any seeds were evaluated. No data was produced to assess the support condition. | next: Fix the matrix_multiply/power-iteration bug (likely a shape mismatch in constructing the adjacency matrix or the column vector) and rerun the full $(n,\alpha)$ grid.

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## Fiedler Participation Entropy Lower-Bounds Tseitin Tree-Resolution Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Anderson localization theory on finite graphs — Rényi-2 participation entropy / inverse participation ratio (IPR) of Laplacian eigenvectors (Anderson 1958; Aizenman–Molchanov 1993; Erdős–Schlein–Yau 2009; Bauerschmidt–Knowles–Yau 2017 'Local Kesten–McKay law and complete delocalization for random regular graphs'; He–Knowles 2021). The functional $IPR(G) := \Sigma_v \varphi_2(v)^4$ of the unit Fiedler eigenvector $\varphi_2$ of $L_G$ is an $L^{4/L}2$ spectral statistic measuring eigenmode spatial concentration; it uses real fourth-power arithmetic and admits NO polynomial $F_q$ extension (since $|\cdot|^4$ vs $|\cdot|^2$ ratios are not preserved under polynomial ring lifts), Aaronson–Wigderson algebrization-safe. Distinct from blacklisted Tseitin attempts: signed-Laplacian holonomy is the gauge-invariant smallest eigenvalue of $L^\sigma$ on a Z/2 line bundle; effective resistance is a pseudoinverse diagonal pairing; Szegedy phase gap is a singular value on the edge-orientation space; p=3 modulus is a Hölder flow-cost program; none is the $L^4$ mass of a Laplacian eigenvector. arXiv 'participation ratio' AND ('Tseitin' OR 'resolution refutation' OR 'proof complexity') returns 0 direct hits with <5 adjacent papers on Anderson delocalization on random regular graphs. × Tree-like Resolution / DPLL refutation size t*(T(G,ω)) for Tseitin XOR formulas T(G,ω) on connected 3-regular graphs G with odd charge ω : V(G) → F_2 (Urquhart 1987; Ben-Sasson–Wigderson 2001; Alekhnovich–Razborov 2001 expansion regime), in the Cook–Reckhow–Krajíček bounded-arithmetic framework where ¬T(G,ω) is Σ^b_0 and log_2 t*(T(G,ω)) lower-bounds $V^0$-witnessing complexity.
- **Recorded**: 2026-05-18 13:00 UTC
- **Entry ID**: `4a5a7ca02658`

### Statement

For every connected 3-regular graph $G$ on $n$ vertices and odd charge $\omega : V(G) \to F_2$, let $\lambda_2(G)$ be the second-smallest eigenvalue of the combinatorial Laplacian $L_G$ and let $\varphi_2$ be a unit eigenvector for $\lambda_2$; define the Fiedler IPR by $IPR(G) := \Sigma_v \varphi_2(v)^4$ (in the degenerate case, minimum over unit eigenvectors of the $\lambda_2$-eigenspace). We conjecture that there is an absolute constant $c > 0$ such that for every such $(G,\omega)$, *$\log_2 t(T(G,\omega)) \geq c \cdot \lambda_2(G) / IPR(G)$, where t*(T(G,ω)) is the minimum tree-like Resolution refutation size. A single $(G,\omega)$ with $\log_2 t^*(T(G,\omega)) < 0.05 \cdot \lambda_2(G)/IPR(G)$ refutes the conjecture.

### Rationale

Ben-Sasson–Wigderson Tseitin lower bounds rest on edge-expansion $h(G)$; Cheeger ties $h$ to $\lambda_2$, but the standard $\sqrt{2\lambda_2} \geq h$ bound is loose unless the Fiedler vector is GLOBALLY DELOCALIZED — Anderson-style — across many vertices. Localization (small $1/IPR \approx O(1)$) signals a low-frequency mode concentrated on a small cut that DPLL can exploit by branching on the bottleneck edges; multiplying $\lambda_2$ by the effective support $1/IPR$ sharpens the spectral component to count how 'global' the Fiedler mode actually is. The invariant is structural on $(G,\omega)$ (two Tseitin formulas both ≡ FALSE can have very different $IPR\cdot\lambda_2$), shielding from Razborov–Rudich, and uses non-ring real arithmetic, shielding from algebrization.

### Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [0211247v1] Inverse spectral problems for Sturm-Liouville operators with singular potentials - [1010.0094v1] An Inverse Spectral Theorem

#### Empirical Test

- exit code: 1, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7da3a703.py", line 213, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7da3a703.py", line 193, in run_trial
    lambda_2, IPR = compute_fiedler_info(L)
                    ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7da3a703.py", line 126, in compute_fied
    eigenvalues, eigenvectors = matrix_eigen(L)
                                ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7da3a703.py", line 111, in matrix_eigen
    eigenvalue = sum(Ax[i] * x[i] for i in range(n))
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7da3a703.py", line 111, in <genexpr>
    eigenvalue = sum(Ax[i] * x[i] for i in range(n))
                     ~~~~~~^~~~~~
TypeError: can't multiply sequence by non-int of type 'float'
```

#### Judge reasoning

The test crashed with a TypeError in matrix_eigen before any trials completed, so no R values were produced and the pre-registered support condition cannot be evaluated. | next: Fix the matrix_eigen bug (Ax appears to be a list of sequences rather than floats; ensure proper matrix-vector multiplication returns a flat numeric vector) and rerun all 900 trials.

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## Hypercontractive Term-Pair Contraction Defect Bounds Monotone CLIQUE DNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier-analytic hypercontractivity — Bonami–Beckner noise kernel applied SYNTACTICALLY to the symmetric-difference distribution of TERM-SUPPORT pairs of a monotone DNF, in the Kahn–Kalai–Linial / Friedgut / Bonami / Mossel tradition. The invariant is a Rényi-2-to-Rényi-4-style log-ratio of the noise-mass D_ρ(F) := (1/s²)·Σ_{i,j} ρ^{|T_i Δ T_j|} at two scales ρ ∈ {1/2, 1/4} — a non-ring functional (uses |·| on integer-valued symmetric differences and log-ratio of exponential sums, with no polynomial F_q extension since ρ^t is real and the symdiff cardinality is not preserved under ring lifts), so the bridge is Aaronson–Wigderson algebrization-safe. Distinct from blacklisted Bonami 4/2 ratio of the centered CLAUSE-INDICATOR polynomial Q_F on DPLL (that was a truth-table-flavored single polynomial, not a term-pair kernel) and from Hypercontractive Term-Support Stable Rank σ(K)=(Σλ)²/Σλ² (Gram-spectrum invariant via eigendecomposition, which crashed in dense matrix code; here we never form eigenvalues — only the scalar log-ratio of two noise-kernel sums). An arXiv search 'Bonami-Beckner' AND ('monotone DNF' OR 'CLIQUE indicator') applied as a TERM-PAIR symdiff log-ratio returns 0 direct hits with <5 adjacent papers (all on truth-table-side hypercontractivity, never on the DNF's term family). × Monotone DNF leaf-size L(F) := Σ_i |T_i| for the k-CLIQUE indicator on K_v with k = 3 fixed and n = v(v−1)/2

edge variables, in the Razborov 1985 / Andreev / Karchmer–Wigderson MONOTONE_CLIQUE regime. The invariant is SYNTACTIC on the DNF (two DNFs with the same Boolean function can have very different η), shielding it from the Razborov–Rudich natural-proofs barrier.

- **Recorded**: 2026-05-18 13:47 UTC
- **Entry ID**: `684e59090c85`

### Statement

For monotone DNF F = ∨*{i=1}^s T_i on n variables (T_i ⊆ [n], s ≥ 2), define the Bonami noise mass D_ρ(F) := (1/s²)·Σ*{i,j=1}^s ρ^{|T_i Δ T_j|} for ρ ∈ (0,1), and the hypercontractive contraction defect η(F) := log_2 D_{1/2}(F) − 2·log_2 D_{1/4}(F). Conjecture: there exist absolute constants C_1 ≤ 6 and c_2 ≥ 0.10 such that (i) for every monotone DNF F with leaf-size L, η(F) ≤ C_1·log_2(L + 2), and (ii) for the canonical minimal 3-CLIQUE DNF F^*_v on K_v (one term per triangle, |T_i| = 3), η(F^*_v) ≥ c_2·√v for all v ≥ 5. A single F violating (i) with margin > 0.5 or a single v ≥ 5 with η(F^*_v) < c_2·√v − 0.05 refutes the conjecture.

### Rationale

D_ρ measures how clustered the term family is in Hamming geometry; the 1/2-vs-1/4 log-ratio η(F) is a discrete analogue of a Bonami 2→4 hypercontractive contraction taken on the term-pair empirical measure, isolating the 'shape' of the symmetric-difference distribution beyond its mean. For poly-leaf-size DNFs the term family is small/sparse, forcing the symdiff distribution to concentrate at large values and bounding η by O(log L); for the 3-CLIQUE family the C(v,3) triangles share a rigid combinatorial overlap pattern (edge-sharing pairs contribute symdiff 4, otherwise 6), creating a two-mode mixture whose mass at the small-symdiff mode scales like 1/v, producing a predictable √v growth in the 2/4 contraction ratio. Because η depends on the multiset of term-supports (not the truth table), it is shielded from Razborov–Rudich, and because it uses only |·| and exponential sums it does not algebrize.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f7d2086c.py", line 98, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f7d2086c.py", line 59, in run_trial
    s_values = [n * n, 2 * n * n, n * n * n]  # n^{1.5} is n^3 in this context
                ^
UnboundLocalError: cannot access local variable 'n' where it is not associated with a value
```

### Judge reasoning

Test crashed due to unbound local variable 'n' before producing results | next: Debug test code to resolve UnboundLocalError and ensure proper variable initialization

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## Lovász-Theta of Term-Conflict Graph Bounds Monotone k-CLIQUE DNF

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Lovász theta / degree-2 Sum-of-Squares SDP relaxation, accessed through the Hoffman ratio bound $\alpha\_H(G) := -\lambda\_min \cdot |V|/(\lambda\_max - \lambda\_min)$ on independence number (Lovász 1979 'On the Shannon capacity of a graph'; Hoffman 1970; Schrijver 1979; Goemans–Williamson 1995). This is the canonical degree-2 SOS object: $\vartheta(G) = \max \Sigma x\_i$ s.t. positive-semidefinite X with $X\_\{ii\}=x\_i$ and $X\_\{ij\}=0$ on edges. Applied STRUCTURALLY to the term-conflict graph G_F of a monotone DNF F (a polymatroid-expansion object on the term family), the invariant is a property of the DNF wiring not of its truth table — two monotone DNFs with the same Boolean function can have wildly different G_F, shielding from Razborov–Rudich natural proofs. The functional uses real eigenvalue extraction and ratio (no F_q polynomial extension exists, since the spectral ratio $\lambda\_min/(\lambda\_max-\lambda\_min)$ is not preserved under polynomial ring lifts), Aaronson–Wigderson algebrization-safe. Distinct from blacklisted Bonami–Beckner stable-rank / term-pair noise-ratio (those use a hypercontractive Gram kernel, not an SDP on the conflict graph), Cauchy-mean-width of the minterm hull (convex-geometry intrinsic 1-volume), Forman–Ricci curvature (Bochner edge-vertex sum), and F_2-corank of minterm incidence (binary-matroid representability). An arXiv search 'Lovász theta' AND ('monotone DNF' OR 'k-CLIQUE indicator' OR 'term-conflict graph') returns 0 direct hits and <5 adjacent papers (Lee–Schreiber–Theis theta-rank, never on monotone DNF term families). × Monotone DNF leaf size $L(F) = \Sigma\_i |T\_i|$ for the k-CLIQUE indicator F*_v on K_v with $k = \lceil \log\_2 v \rceil$ and $n = v(v-1)/2$ edge variables (Razborov 1985 / Andreev / Karchmer–Wigderson regime). The k=log v scaling makes the canonical minterm DNF of size (v choose k) super-polynomial in n, the standard MONOTONE_CLIQUE focus's stepping stone toward the Razborov $n^{\Omega(k}${1/4}) monotone lower bound and a V^0 separation via Buss–Pudlák–Krajíček-style approximator counting.
- **Recorded**: 2026-05-18 16:26 UTC
- **Entry ID**: `6227dd6720d4`

### Statement

For a monotone DNF $F = \vee\_\{i=1\}^s T\_i$ on n variables, define the term-conflict graph G_F on vertex set [s] with edges $E(G\_F) = \{\{i,j\} : T\_i \cap T\_j \neq \emptyset\}$, let A_F be its {0,1}-adjacency matrix with extreme eigenvalues $\lambda\_1 \geq \lambda\_s$, and set $\alpha\_H(F) := -\lambda\_s \cdot s / (\lambda\_1 - \lambda\_s)$ if $\lambda\_1 > 0$ else $\alpha\_H(F) := s$. Define the measure $\mu(F) := s / (\alpha\_H(F)\cdot(1 + \log\_2(1 + \max\_i |T\_i|)))$. Conjecture: (i) for every two monotone DNFs F,G on disjoint term-lists, $\mu(F \vee G) \leq \mu(F) + \mu(G)$ and $\mu(F \wedge G) \leq \mu(F) + \mu(G)$ (where $F \wedge G$'s term list is {T_i ∪ T'*j}*{i,j}}); (ii) for every monotone DNF F with at most $s \leq poly(n)$ terms and max term size $\leq C\cdot\log n$, $\mu(F) \leq C'\cdot\log\_2 n$ with absolute $C' > 0$; (iii) for the k-CLIQUE indicator F*_v on K_v with $k = \lceil\log\_2 v\rceil$, expressed as the canonical disjunction of k-clique minterms in edge-variables, $\mu(F^*\_v) \geq c\cdot v$ for an absolute constant $c > 0$. A single random-poly-size DNF with $\mu > 8\cdot\log\_2 n$, or a k-CLIQUE canonical DNF F*_v with $\mu(F^*\_v) < v/16$, refutes.

### Rationale

Lovász theta and its Hoffman spectral surrogate are the canonical degree-2 SOS lift of independence number; applied to the term-conflict graph of a monotone DNF they probe the polymatroid expansion of the term family — exactly the quantity that Razborov-style approximators try to capture but on the SYNTACTIC side. The k-CLIQUE canonical DNF has terms = k-cliques in K_v, inducing a dense Johnson/Kneser-type conflict graph whose spectrum is highly concentrated ($\lambda\_1 \approx k\cdot(v-k)\cdot C$, $\lambda\_s \approx -\Theta(k)$), so $\alpha\_H \approx s/(v-k)$ and μ scales linearly in v. Generic poly-size monotone DNFs with short terms have low-density conflict graphs with $\alpha\_H \approx s/(\text{max-degree}+1)$, keeping μ at log-scale; the bridge therefore exploits the SOS-2 spectral fingerprint of polymatroidal term overlap without ever touching the truth table.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.15s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6bb3cd60.py", line 160, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6bb3cd60.py", line 116, in run_trial
    mu = compute_mu(terms, adj_matrix)
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6bb3cd60.py", line 86, in compute_mu
    lambda_max, lambda_min = compute_extreme_eigenvalues(adj_matrix)
                             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6bb3cd60.py", line 59, in compute_extre
    lambda_max, _ = power_iteration(M)
                    ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6bb3cd60.py", line 37, in power_iterati
    b_k_norm = matrix_norm(b_k)
               ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6bb3cd60.py", line 30, in matrix_norm
    return math.sqrt(sum(sum(x**2 for x in row) for row in M))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6bb3cd60.py", line 30, in <genexpr>
    return math.sqrt(sum(sum(x**2 for x in row) for row in M))
                            ^^^^^^^^^^^^^^^^^^^
TypeError: 'float' object is not iterable
```

**Judge reasoning**

The test crashed with a TypeError in the power_iteration routine (matrix_norm received a 1D vector instead of a 2D matrix), producing no data on either the random-DNF or k-CLIQUE criteria. No pre-registered condition can be evaluated. | next: Fix matrix_norm to handle 1D vectors (or use numpy.linalg.eigvalsh for the symmetric {0,1}-adjacency matrix) and rerun the 30-seed protocol.

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## Sidon B_2-Witness Failure of ACC⁰[2] for Sipser via MOD-Gate Outputs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sidon / B_2-set additive combinatorics — the Erdős–Turán maximum representation function r*(A) := max_{w ∈ A+A} #{(a,a') ∈ A² : a + a' = w}, classifying A ⊆ F_2^N as a Sidon (B_2) set iff* r(A) = 2 (only diagonal pairs), with the 'Sidon defect' sd(A) := r*(A) − 2 ∈ [0, |A|−2] measuring deviation from B_2-ness (Erdős–Turán 1941 'On a problem of Sidon in additive number theory'; Lindström 1969 'An inequality for B_2-sequences'; Cilleruelo 2010; O'Bryant 2004 survey 'A complete annotated bibliography of work related to Sidon sets and B_h sets'). A non-ring max-of-integer-counts functional: uses max + integer counting + F_2-XOR; the OUTER max is sup-style and no F_q polynomial extension preserves max-representation count under ring lifts, so the invariant is Aaronson–Wigderson algebrization-safe. Distinct from blacklisted Plünnecke doubling |A+A|/|A| on truth-table top-Fourier supports (a sumset CARDINALITY functional), Chang η-spectrum cardinality on wire-source labels (a Fourier-spectrum functional), additive energy E(A)=Σ_w r(A,w)² on out-degree sequences (an L² functional in Z, not max in F_2^N), Lehmer-pair close-zero spacings, and Bourgain–Tzafriri subcolumn dispersion;* r is the MAX (not sum, not size) of the representation function on F_2^N gate-output vectors. ArXiv 'Sidon set' AND ('ACC^0' OR 'lifting theorem' OR 'circuit lower bound') returns 0 direct hits with <5 adjacent papers (Behrend-Roth, Ruzsa-Plünnecke), none on ACC⁰ MOD-gate output multisets. × ACC⁰[2] circuit size s(C) and depth d(C) for the depth-d Sipser–Yao function S_{n,d} (alternating AND/OR tree of fan-in n^{1/d}), in the Williams 2014 NEXP⊄ACC⁰ / Murray–Williams 2018 algorithmic-method regime targeting explicit ACC⁰ lower bounds. Bridged to LIFTING via 'within-circuit gate-output lifting' in the Raz–McKenzie

/ Göös–Pitassi–Watson tradition: each MOD_2 gate g is an F_2-linear functional of its inputs, and its truth-table on N=1024 seed-fixed random inputs lifts the circuit DAG into an additive-combinatorial object V_C ⊆ F_2^N. The invariant sd is STRUCTURAL on the circuit wiring (two $ACC^0$[2] circuits with identical Boolean function can have different sd), shielding from Razborov–Rudich natural proofs; the within-circuit lifting plays the role of the standard query-to-communication gadget composition.
- **Recorded**: 2026-05-18 17:00 UTC
- **Entry ID**: `b6fc094b0a25`

### Statement

Let C be an $ACC^0$[2] circuit on n inputs of depth d with m_C MOD_2 gates of fan-in 4 whose output vectors are F_2-linearly independent. Fix N=1024 uniformly random inputs (seed s), let v_g ∈ F_2^N be gate g's output vector, and V_C(s) := {v_g : g a MOD_2 gate of C}. Define r*(C,s) := max_{w ∈ F_2^N} #{(a,b) ∈ V_C(s)² : a XOR b = w} and sd(C) := median over 30 seeds of (r*(C,s) − 2). Conjecture (B_2-Witness for $ACC^0$[2]): for every such $ACC^0$[2] circuit C with m_C ≥ 8 and d ≤ 5, sd(C) ≤ ⌈√m_C⌉, i.e., MOD_2 gate outputs are approximately-Sidon under typical input subsamples. A single (C, seed-batch) with sd(C) > ⌈√m_C⌉ refutes the conjecture; corollary: by Razborov–Smolensky, the Sipser function S_{n,d} forces a non-random F_2-polynomial structure in V_C breaking the bound, separating S_{n,d} from depth-d $ACC^0$[2].

### Rationale

$ACC^0$[2] circuits of bounded depth compute low-degree F_2-polynomial approximations of their inputs (Razborov 1987; Smolensky 1987), so MOD_2 gate outputs span an F_2-vector subspace of dimension Θ(n^{O(d)}) inside F_2^N which, for m_C □ √(dim), behaves like a random Sidon set with r* concentrating at 2. The Sipser function's depth-d alternation forbids such randomness: any $ACC^0$[2] computation of S_{n,d} must inject correlations that drive r* above √m_C, yielding an $ACC^0$ obstruction. The conjecture is shielded from Razborov–Rudich (sd is wiring-dependent, not truth-table dependent), from Aaronson–Wigderson (max + count are non-ring), and instantiates the LIFTING focus by treating MOD-gate output vectors as a gadget-style algebraic lift of the underlying decision-tree structure.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66924d2a.py", line 165, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66924d2a.py", line 116, in run_trial
    gate_outputs.append(evaluate_circuit(circuit, input_vec))
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66924d2a.py", line 88, in evaluate_circ
    gate_input = [inputs[i] ^ negations[i] for i in inputs_idx]
                              ~~~~~~~~~^^^
IndexError: list index out of range
```

### Judge reasoning

The test crashed with an IndexError in evaluate_circuit before producing any RESULT line, so no data was collected against the pre-registered criterion. | next: Fix the indexing bug in evaluate_circuit (ensure negations array is sized to match the circuit's input fan-in / gate inputs_idx range) and rerun the 810-circuit sweep.

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## RSK Shape Concentration Separates Permanent from Padded Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Robinson-Schensted-Knuth combinatorics and Plancherel measure on Young diagrams (Schensted 1961; Logan-Shepp 1977; Vershik-Kerov 1977; Baik-Deift-Johansson 1999; Romik 2015 'The Surprising Mathematics of Longest Increasing Subsequences'). The RSK shape map σ ↦ λ(σ) ⊢ n is computed by patience-sort / row insertion — a comparison-and-bumping algorithm with NO polynomial extension over F_q, since longest-increasing-subsequence length is a max/sort functional that does not algebrize per Aaronson-Wigderson. Distinct from blacklisted Specht-isotype mass (which projects c(f) onto λ-isotypes via character sums Σ_σ χ^λ(σ)c_σ) — here we PARTITION S_n by RS shape and sum $|c\_σ|^2$ fiberwise, never invoking χ^λ; distinct from Eulerian-descent entropy (a Worpitzky descent-count binning, not an LIS shape); distinct from Patience-Sort LIS XOR-lifted (different target object: XOR-lifted CC matrices on F_2^n, not the GCT permutation-coefficient vector of perm vs padded-det). An arXiv search 'RSK shape distribution' AND ('determinantal complexity' OR 'GCT' OR 'permanent versus determinant' OR 'orbit closure') returns 0 direct hits with <5 adjacent papers, all on Plancherel asymptotics in random-matrix theory. × Mulmuley-Sohoni Geometric Complexity Theory: determinantal complexity dc(perm_n), the smallest m such that perm_n ∈ GL_{$n^2$}·orbit-closure of □(y)^{n-m}·det_m(L(y)) for □ a linear form and L an m×m matrix of linear forms in the $n^2$ matrix variables y={y_{ij}} (Mignon-Ressayre 2004 $n^2/2$ lower bound; Landsberg 2017; Burgisser-Ikenmeyer 2013). The permutation-coefficient vector c(f) ∈ Z^{n!} with c_σ(f) := coeff of ∏*i x*{i,σ(i)} in f gives a canonical S_n × S_n-equivariant structural fingerprint that is preserved under row/column-permutation substitutions (a subgroup of GL_{$n^2$}), suitable as a partial orbit-closure obstruction in the canonical det-vs-perm separation testbed.
- **Recorded**: 2026-05-18 17:28 UTC
- **Entry ID**: `c0170ef261da`

### Statement

For a homogeneous polynomial f of degree n in matrix variables {x_{ij}}*{1≤i,j≤n}, let c_σ(f) := coeff of* ∏i x*{i,σ(i)} in f, define the RSK-shape weighted distribution P_f(λ) := (Σ*{σ:λ(σ)=λ} $c\_σ(f)^2$)/(Σ_σ $c\_σ(f)^2$) when the denominator is nonzero, and set K(f) := max_{λ⊢n} P_f(λ). Conjecture: there exist absolute constants $n_0$, C such that for every n ≥ $n_0$ and every padded determinant g = □^{n-m}·det_m(L) with m ≤ n-1, □ a nonzero linear form, L an m×m matrix of linear forms over Q in {x_{ij}}, and Σ_σ $c\_σ(g)^2$ > 0, the inequality K(g) ≥ C·K(perm_n) holds with C = 2 and $n_0$ = 5. A single (n,m,□,L) tuple at n ∈ {5,6,7,8} with K(g) < 2·K(perm_n) and Σ_σ $c\_σ(g)^2$ > 0 falsifies the conjecture.

### Rationale

The all-ones coefficient vector of perm_n drives P_{perm_n} to be EXACTLY the Plancherel measure $(f^λ)^2/n!$, whose maximum at the Vershik-Kerov limit shape decays as O(n^{-1/4}) — perm_n is maximally RS-diffuse. Padded determinants, by contrast, have multilinear-permutation coefficients controlled by sgn(ρ)·signed-permanent-cofactor sums over only n!/(m!·(n-m)!)-many σ-supports, concentrating $|c\_σ(g)|^2$ on permutations factoring through a fixed m-block — these factorings live in a small union of (P,Q)-shape fibers, forcing K(g) ≥ constant. The bridge therefore predicts a quantitative diffuseness gap that is structural on (□,L) (two padded forms with identical Boolean truth tables can have different K), shielding from Razborov-Rudich; and the RSK map uses only comparisons, shielding from Aaronson-Wigderson algebrization.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_613d9fe5.py", line 126, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_613d9fe5.py", line 95, in run_trial
    c_padded_det = compute_c_padded_det(n, m, l, L, seed)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_613d9fe5.py", line 86, in compute_c_pad
    c[sigma] = term
    ~^^^^^^^
TypeError: list indices must be integers or slices, not tuple
```

### Judge reasoning

The test crashed with a TypeError (list indexed by tuple) in compute_c_padded_det before any seeds produced data, so neither the support nor falsification condition could be evaluated. | next: Fix the coefficient storage to use a dict keyed by permutation tuples (or convert sigma to a flat index) and rerun the pre-registered 30-seed sweep over (n,m) ∈ {5,6,7,8}×{1,2,3}.

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## Tusnady 2-Box Discrepancy of Clause-Polarity Cloud Bounds DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric combinatorial discrepancy — Tusnady's problem and the Chazelle–Matousek axis-aligned 2-box range discrepancy of finite point sets in R^d (Beck 1981; Roth 1954; Schmidt 1972; Matousek 'Geometric Discrepancy' 1999; Chazelle 'The Discrepancy Method' 2000). The 2-marginal box discrepancy is computed by enumerating coordinate pairs and sign quadrants — an O(n^2 m) combinatorial functional using only |·| and max (non-ring; no F_q polynomial extension exists since max-of-integer-deviations is not preserved under polynomial ring lifts), so the bridge is Aaronson–Wigderson algebrization-safe. An arXiv search 'Tusnady discrepancy' AND ('DPLL' OR 'tree-Resolution' OR 'CNF refutation' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers (Chen–Matousek–Skriganov on Tusnady–Beck lower bounds, never on CNF refutations). Distinct from blacklisted Bourgain–Tzafriri sub-column dispersion (column-subset operator norms on sign matrices), Halasz L^2 spectrum discrepancy (Laplacian-spectrum CDF deviation), Beck-style signed-support L_inf discrepancy on incidence rows, and Mostar/Forman-Ricci structural-graph invariants — none of which is an axis-aligned 2-box range discrepancy on the clause-polarity point cloud. × Tree-like Resolution / DPLL refutation size t*(F) for random unsatisfiable 3-CNFs F at clause density alpha in [4.0, 5.0] (Chvatal–Szemeredi 1988; Beame–Karp–Pitassi–Saks 2002; Ben-Sasson–Wigderson 2001; Achlioptas–Beame–Molloy 2004), in the Cook–Reckhow–Krajicek bounded-arithmetic regime where ¬F is Sigma^b_0 and log_2 t*(F) controls V^0-witnessing complexity. The invariant D_2(F) is STRUCTURAL on the clause LIST (multiplicities, polarities), so two unsat 3-CNFs with identical satisfiability function (both ≡ FALSE) can have very different D_2(F), shielding the invariant from the Razborov–Rudich natural-proofs barrier.

- **Recorded**: 2026-05-18 17:45 UTC
- **Entry ID**: `d8440effcf4b`

### Statement

For an unsatisfiable 3-CNF F with m clauses on n variables, define the 2-marginal polarity discrepancy D_2(F) := max over 1 ≤ i < j ≤ n and (s,t) in {-,+}^2 of |c_{ij}^{st}(F) − mu_{ij}^{st}|, where c_{ij}^{st}(F) is the number of clauses of F containing literal x_i^s AND literal x_j^t, and mu_{ij}^{st} = m·(3/n)·(2/(n−1))·(1/4) is the expected count under the uniform random 3-CNF model with parameters (m,n). Conjecture: for unsat 3-CNFs F drawn at clause density alpha = 4.5 on n in {12, 14, 16, 18, 20} variables, the Spearman rank correlation between −D_2(F)/sqrt(m) and log_2 t*(F) across 30 seeds × 5 sizes (150 instances) is ≥ +0.30 with two-sided p < 0.05. Equivalently, formulas whose clause-polarity cloud is more uniformly spread over axis-aligned 2-box ranges tend to have provably larger DPLL refutation trees, and the inequality is falsified by a single (30 seed × 5 size) ensemble whose empirical correlation falls below 0.30.

### Rationale

Geometric discrepancy quantifies how evenly a point configuration is spread over axis-aligned ranges; CNF hardness for DPLL is widely conjectured to track ‘uniformity’ of clause structure (Tseitin on expanders and random alpha = 4.267 formulas both exhibit balanced literal distributions and yield exponentially large DPLL trees). The 2-marginal box discrepancy is the simplest non-trivial Tusnady-style geometric discrepancy that ignores 1-marginal polarity bias (already known to be a weak heuristic predictor for DPLL via Jeroslow–Wang) but captures the next-order clause-pair polarity correlations relevant to unit-propagation cascades. The bridge proposes ‘spread’ of the clause-polarity cloud as a structural CNF invariant within the Chazelle–Matousek–Bourgain dispersion programme, distinct from spectral, additive-combinatorial, and curvature-based attempts.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `0`, elapsed: `0.07s`


```
.09999999999999998"}
TRIAL: {"seed": 421, "metric_name": "spearman_correlation", "metric_value": 0.19999999999999996, "in
TRIAL: {"seed": 463, "metric_name": "spearman_correlation", "metric_value": 0.9, "instances_tested":
TRIAL: {"seed": 503, "metric_name": "spearman_correlation", "metric_value": 0.9, "instances_tested":
TRIAL: {"seed": 547, "metric_name": "spearman_correlation", "metric_value": 0.7, "instances_tested":
TRIAL: {"seed": 593, "metric_name": "spearman_correlation", "metric_value": 1.0, "instances_tested":
TRIAL: {"seed": 631, "metric_name": "spearman_correlation", "metric_value": 0.4, "instances_tested":
TRIAL: {"seed": 677, "metric_name": "spearman_correlation", "metric_value": 1.0, "instances_tested":
TRIAL: {"seed": 727, "metric_name": "spearman_correlation", "metric_value": 1.0, "instances_tested":
TRIAL: {"seed": 773, "metric_name": "spearman_correlation", "metric_value": 1.0, "instances_tested":
TRIAL: {"seed": 821, "metric_name": "spearman_correlation", "metric_value": 0.7, "instances_tested":
TRIAL: {"seed": 877, "metric_name": "spearman_correlation", "metric_value": -0.5, "instances_tested"
0.5"}
TRIAL: {"seed": 929, "metric_name": "spearman_correlation", "metric_value": 0.9, "instances_tested":
RESULT: FALSIFIED counterexample="correlation=0.19999999999999996" first_failing_seed=37
```


### Judge reasoning

Safety rail: critic_challenge_falsified | original: The test’s RESULT line reports FALSIFIED with counterexample correlation=0.20 at seed=37, and the critic flags that the pre-registered 150-instance Sp | next: Recompute Spearman ρ on the pooled 150-instance set (not per-seed n=5 subsets), partial out the n-trend by within-n ranking or residualizing $\log_2 t^*(F)$ on n before correlating with $-D_2/\sqrt{m}$.

---

## Hook-Length Dim of Tseitin T-Join Partition Lower-Bounds Tree-Res

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symmetric-group Specht-module dimensions via Frame–Robinson–Thrall hook-length formula (Frame–Robinson–Thrall 1954; James–Kerber 1981; Sagan 2001 'The Symmetric Group'). The hook-length functional $\dim S^\lambda = n!/\prod_{c\in\lambda} h(c)$ on integer partitions is a representation-theoretic dimension count from algebraic combinatorics and a canonical GCT-flavored invariant in the Mulmuley–Sohoni / Bürgisser–Ikenmeyer plethysm tradition. Applied here SYNTACTICALLY to the canonical T-join path-length partition $\pi(G,\omega)$ of a charged 3-regular graph, the invariant uses only integer factorials and divisions on a small partition (no ring/$F_q$ polynomial extension — factorials and hook quotients are not preserved under polynomial ring lifts), so the bridge is Aaronson–Wigderson algebrization-safe. The invariant is STRUCTURAL on $(G,\omega)$ (two Tseitin formulas with identical Boolean function ≡ FALSE can have very different $\pi$), shielding from Razborov–Rudich natural-proofs. An arXiv search for 'hook length formula' AND ('Tseitin' OR 'resolution refutation' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers (all Specht/plethysm work in GCT or on permutation patterns, never on charged 3-regular Tseitin partitions). Distinct from blacklisted Specht-isotype mass on perm-vs-det $c_\sigma$ vectors (an isotype projection $\chi^\lambda \cdot c(f)$ on the permutation-monomial coefficient vector of homogeneous polynomials, NOT a hook product on a graph-theoretic partition), distinct from Eulerian-descent entropy / Tamari-Pallo rank / Patience-sort LIS / RSK shape concentration, and distinct from all prior Tseitin-side spectral, modulus, resistance, holonomy, Szegedy, fatgraph, and curvature attempts. × Tree-like Resolution / DPLL refutation size t*(T(G,ω)) for Tseitin XOR formulas on connected 3-regular graphs G with odd charge $\omega : V(G) \to F_2$ (Urquhart 1987; Ben-Sasson–Wigderson 2001; Alekhnovich–Razborov 2001 expansion regime), in the Cook–Reckhow–Krajíček bounded-arithmetic framework where* $\neg T(G,\omega)$ *is* $\Sigma^b_0$ *and* $\log_2 t$ lower-bounds $V^0$-witnessing complexity — the canonical Frege-depth / Extended-Frege stepping stone targeted by the PROOF_COMPLEXITY_TSEITIN approach.
- **Recorded**: 2026-05-18 18:40 UTC
- **Entry ID**: `a677f9fc0b8d`

### Statement

For every connected 3-regular graph G on n vertices with odd charge $\omega : V(G) \to F_2$, define the T-join path partition $\pi(G,\omega)$ as the multiset of edge-lengths of the canonical greedy minimum-weight T-join decomposition (T = odd-charge vertices, parity-fixed by appending the lex-smallest non-charged vertex when |T| is odd; pairing is greedy BFS-closest pairs by lex tie-break), and let $\rho(G,\omega) := \log_2(\dim S^{\pi(G,\omega)})$ with $\dim S^{\pi(G,\omega)}$ computed by the Frame–Robinson–Thrall hook-length formula on $\pi$. Then there exists a universal constant $c \geq 0.15$ such that for every $(G,\omega)$ with $|E(G)| \leq 40$ the tree-Resolution refutation size t*(T(G,ω)) satisfies log_2 t*$(T(G,\omega)) \geq c \cdot \rho(G,\omega) / \log_2(n+2)$. One counterexample $(G,\omega)$ with the inequality violated falsifies the conjecture.

### Rationale

Tseitin tree-Resolution lower bounds in the Ben-Sasson–Wigderson framework are controlled by how the odd charge spreads through the graph: dispersed charges force long path obligations that resolution must witness. The T-join path partition $\pi(G,\omega)$ captures exactly this spread, and the hook-length dimension $\dim S^\pi$ — a representation-theoretic, GCT-flavored coarsening — concentrates on partitions with many roughly equal parts (which is precisely the regime where Tseitin is provably hard) and collapses for clustered charges (where Tseitin is easy via local cancellation). This bridges the algebraic-combinatorial machinery used in the perm-vs-det / GCT_DET_PERM program (Specht modules, hook formulas, plethysm) with the proof-complexity hardness of expander-Tseitin, an angle no prior literature appears to have explored.

### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.72s`

```
TRIAL: {'metric_name': 'r', 'metric_value': -59.79470570797252, 'instances_tested': 3, 'conjecture_h
1.0, log_t_star=18.0, log_n=3.321928094887362, r=-59.79470570797252'}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0eec2ac0.py", line 188, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0eec2ac0.py", line 139, in run_trial
    adj = generate_3_regular_graph(n, seed)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0eec2ac0.py", line 25, in generate_3_re
    v = random.choice([x for x in stubs if x != u])
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 347, in choice
    raise IndexError('Cannot choose from an empty sequence')
IndexError: Cannot choose from an empty sequence
```

### Judge reasoning

The test crashed with an IndexError in generate_3_regular_graph before completing the pre-registered 120 instances; only 3 instances ran and the reported counterexample has ρ=-1.0, which violates the ρ>0 precondition of the falsification criterion. Neither support nor falsification can be unambiguously established from this run. | next: Fix the 3-regular graph generator (handle the stub-pairing dead-end by restarting or using a configuration-model with retries) and re-run the full 120-instance p

---

## Dismantlability Core of Clause-Sharing Graph Bounds Tree-Res Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Pursuit-evasion / cop-number combinatorics — the Nowakowski–Winkler 1983 / Quilliot 1978 dismantling order on finite graphs, where a vertex v is DOMINATED in G iff there exists u ≠ v with N_G[v] ⊆ N_G[u] (closed-neighborhood inclusion). Iteratively delete a dominated vertex until none remains; the surviving subgraph is the unique-up-to-isomorphism dismantling core K(G), and the DISMANTLABILITY DEFECT Δ(G) := |V(K(G))| measures how far G is from being cop-win (Aigner–Fromme 1984; Bonato–Nowakowski 2011 monograph 'The Game of Cops and Robbers on Graphs'). The functional uses only set-inclusion comparisons on closed neighborhoods (non-ring; no F_q polynomial extension exists since N[v] ⊆ N[u] is a Boolean lattice predicate not a polynomial identity), so the bridge is Aaronson–Wigderson algebrization-safe; the per-vertex domination test uses set comparison and is computable in O(m·deg²). An arXiv search for 'dismantlable graph' AND ('Frege' OR 'tree-Resolution' OR 'DPLL' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers (cop-number parameterized-complexity work, never as a CNF-structural invariant). Distinct from all prior CNF-graph attempts: Mostar (edge distance-asymmetry), Forman–Ricci

(Bochner edge-vertex sum), fatgraph genus (ribbon embedding), independence-complex Euler χ̃ (Stanley–Reisner), effective resistance (pseudoinverse pairing), p=3 modulus (Hölder flow-cost), Viennot heap radius (Cartier–Foata clique polynomial root), cluster-expansion convergence radius (Fernandez–Procacci fixed-point) — none uses the closed-neighborhood domination order or its dismantling core size. × Tree-like Resolution / DPLL refutation size t*(F) for unsatisfiable 3-CNFs F in the Ben-Sasson–Wigderson 2001 / Beame–Pitassi 2001 / Krajíček 1995 expansion regime, where ¬F is Σ^b_0 and log_2 t*(F) controls V^0-witnessing complexity. Tree-Resolution size is the canonical Frege/Extended-Frege depth stepping stone in the Cook–Reckhow–Krajíček bounded-arithmetic framework: a lower bound log_2 t*(F) ≥ φ(F) immediately yields a Frege-depth bound d_F(F) ≥ Ω(φ(F)) via Pudlák–Buss feasible-interpolation simulations, the target route toward super-polynomial EF lower bounds for explicit tautologies.
- **Recorded**: 2026-05-18 19:09 UTC
- **Entry ID**: `596827cbde7c`

### Statement

For every unsatisfiable 3-CNF F with n variables and m clauses, let G_F be the clause-sharing graph (vertices = clauses of F; edge {C, C'} iff vars(C) ∩ vars(C') ≠ ∅), and let Δ(G_F) be its dismantlability defect (vertex count of the dismantling core obtained by iteratively deleting closed-neighborhood-dominated vertices, in any order — well-defined by Nowakowski–Winkler). Conjecture: log_2(t*(F) + 1) ≥ (1/8) · n · Δ(G_F) / m. A single unsat 3-CNF F whose ratio R(F) := 8·m·log_2(t*(F)+1) / (n·max(Δ(G_F),1)) is strictly less than 1 refutes the conjecture.

### Rationale

A clause-sharing graph is dismantlable precisely when the formula admits a sequential local-elimination order under closed-neighborhood domination — exactly the structural property exploited by DPLL's unit-propagation cascades, which yield polynomial-size tree-Resolution proofs for 2-SAT-like and Horn-like CNFs. Non-dismantlable cores correspond to expander-like sub-formulas whose Ben-Sasson–Wigderson width is Ω(n), forcing exponential refutation size. The invariant is STRUCTURAL on the clause list (two semantically identical unsat 3-CNFs with both functions ≡ FALSE can yield different G_F and hence different Δ), shielding it from the Razborov–Rudich natural-proofs barrier; the set-inclusion test is non-arithmetic, shielding it from Aaronson–Wigderson algebrization.

### Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2501.18019v2] An exact closed walks series formula for the complexity of regular graphs and some related bounds - [1508.02426v1] A note on independence complexes of chordal graphs and dismantling - [0911.3482v5] Complexity of Networks (reprise) - [1703.04427v3] Capture-time Extremal Cop-Win Graphs - [1603.04266v2] Cops and Robbers ordinals of cop-win trees

### Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_24db1ff9.py", line 156, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_24db1ff9.py", line 110, in run_trial
    delta = compute_dismantlability_defect(graph)
```

```
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_24db1ff9.py", line 53, in compute_disma
    if all(any(u in graph[v] for u in graph[w]) for w in neighbors):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_24db1ff9.py", line 53, in <genexpr>
    if all(any(u in graph[v] for u in graph[w]) for w in neighbors):
                                    ~~~~~^^^
KeyError: 45
```

#### Judge reasoning

The test crashed with a KeyError in compute_dismantlability_defect before producing any data, so neither the support nor the falsification condition could be evaluated. | next: Fix the dismantlability routine to operate on a mutable copy of the graph (removing deleted vertices from all adjacency lists) so KeyError cannot occur, then rerun the 720-instance sweep.

---

### Baker-Norine ω-Gonality Lower-Bounds Tseitin DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Riemann-Roch / chip-firing divisor theory on finite graphs (Baker-Norine 2007 'Riemann-Roch and Abel-Jacobi theory on a finite graph'; Hladký-Kráľ-Norine 2013 'Rank of divisors on tropical curves'; Cools-Draisma-Payne-Robeva 2012 tropical Brill-Noether; Caporaso 2014). The Baker-Norine rank r_BN(D) of a divisor D ∈ Z^V uses Dhar's burning algorithm — a parallel chip-firing reduction with set-firing on subsets S ⊆ V{base} satisfying D(v) ≥ deg_S(v); the rank test for r ≥ 1 enumerates 'D−v ~ effective' for every v ∈ V. The functional uses integer subtraction, max/argmax over firing sets, and the order predicate D ≥ 0 (non-ring; there is no F_q polynomial extension since '≥ 0' and base-reduced canonical form are not preserved under polynomial ring lifts), so the bridge is Aaronson-Wigderson algebrization-safe. ArXiv 'Baker-Norine' AND ('Tseitin' OR 'resolution refutation' OR 'proof complexity') returns 0 direct hits with <5 adjacent papers (all on tropical Brill-Noether, never on CNF refutation). Distinct from blacklisted 2-Sylow rank of K(G) (a property of G alone, here we use rank of a (G,ω)-specific divisor), hook-length dim of T-join partition (representation-theoretic dimension of a Specht module), signed-Laplacian holonomy / effective resistance / p=3 modulus / Szegedy phase gap / Fiedler IPR (spectral/metric), fatgraph genus (oriented surface embedding), and dismantlability core (closed-neighborhood domination order) — none uses Baker-Norine divisor rank of the charge divisor. × Tree-like Resolution / DPLL refutation size t*(T(G,ω)) for Tseitin XOR formulas T(G,ω) on connected 3-regular graphs G with odd charge ω: V → F_2 (Urquhart 1987; Ben-Sasson-Wigderson 2001; Alekhnovich-Razborov 2001 expansion regime), in the Cook-Reckhow-Krajíček bounded-arithmetic framework where ¬T(G,ω) is Σ^b_0 and log_2 t*(T(G,ω)) lower-bounds V^0-witnessing complexity. Tree-Resolution size is the canonical Frege-depth stepping stone via Pudlák-Buss feasible interpolation; Tseitin lower bounds for Extended Frege remain an open target.
- **Recorded**: 2026-05-18 19:38 UTC
- **Entry ID**: `9dfcbea8b0a8`

#### Statement

Let G be a connected 3-regular graph on n vertices with odd charge ω: V → F_2 (Σω ≡ 1 mod 2), and let T = ω^{−1}(1). Define the ω-gonality ρ(G,ω) := min{deg(D) : D ∈ Z^V, D ≥ 0, supp(D) ⊆ T, r_BN(D) ≥ 1}, where r_BN is the Baker-Norine rank computed via Dhar's burning algorithm (D has rank ≥ 1 iff D − v is linearly equivalent to an effective divisor for every v ∈ V). Then for every such (G,ω) with n ≥ 8, log_2 t*(T(G,ω)) ≥ ρ(G,ω) / 4, where t*(T(G,ω)) is the smallest tree-like Resolution refutation size of the Tseitin XOR CNF.

### Rationale

The Baker-Norine rank of the charge divisor measures whether the odd-vertex set T can support a non-trivial linear system on the chip-firing Jacobian of G — a global divisor-theoretic obstruction analogous to the parity obstruction that forces Tseitin to be unsatisfiable, but quantitative rather than binary. High ω-gonality means no small effective divisor on T 'moves' under any single chip removal, witnessing a structural rigidity of (G,ω) that should propagate to refutation difficulty via the Ben-Sasson–Wigderson width-vs-expansion lemma applied to the chip-firing-induced cycle-space decomposition. The invariant is structural on (G,ω) (two formulas with identical constant-FALSE truth tables have different ρ), shielding from Razborov-Rudich; Dhar's burning uses non-ring max/order operations safe from Aaronson-Wigderson.

### Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1504.06713v2] Computing graph gonality is hard - [2010.07992v2] Ternary and Quaternary Curves of Small Fixed Genus and Gonality with Many Rational Points - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1407.6958v3] Chip-firing games on Eulerian digraphs and NP-hardness of computing the rank of a divisor on a graph

### Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1474ec5.py", line 184, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1474ec5.py", line 158, in run_trial
    t_star = compute_t_star(adj, omega)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1474ec5.py", line 124, in compute_t_st
    t_star = dpll(clauses, {})
             ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1474ec5.py", line 122, in dpll
    return dpll([c for c in clauses if lit not in c], {**assignment, lit: True}) + \
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1474ec5.py", line 122, in dpll
    return dpll([c for c in clauses if lit not in c], {**assignment, lit: True}) + \
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1474ec5.py", line 122, in dpll
    return dpll([c for c in clauses if lit not in c], {**assignment, lit: True}) + \
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 6 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1474ec5.py", line 123, in dpll
    dpll([c for c in clauses if (-lit,) not in c], {**assignment, lit: False})
                                ^^^^
TypeError: bad operand type for unary -: 'tuple'
```

### Judge reasoning

The test crashed with a TypeError ('bad operand type for unary -: tuple') inside the DPLL routine before any trials produced data, so neither the violation count nor the Spearman correlation could be evaluated. | next: Fix the DPLL implementation to represent clauses as sets/lists of signed integer literals (not tuples) so that negation works, then re-run the 90-trial protocol.

---

## Morse-Hedlund Factor Complexity of XOR-Lifted Rows Upper-Bounds D(f)

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symbolic dynamics — Morse–Hedlund factor (subword) complexity p_s(k)=#{distinct length-k contiguous substrings of word s}, in the Morse–Hedlund 1938 'Symbolic dynamics II: Sturmian trajectories' / Cassaigne 1997 / Allouche–Shallit 'Automatic Sequences' / Berstel–Lauve–Reutenauer–Saliola 'Combinatorics on Words' tradition. The functional uses string equality and hashing only (no F_q polynomial extension preserves substring multiplicity), so the bridge is Aaronson–Wigderson algebrization-safe. An arXiv search 'subword complexity' AND ('lifting theorem' OR 'communication complexity' OR 'XOR-lifted matrix') returns 0 direct hits with <5 adjacent papers (all on Sturmian/automatic-sequence number theory). Distinct from blacklisted Patience-Sort LIS (a sort-based extremal length), RSK shape (Plancherel partition), Tamari–Pallo rank (Catalan tree order), and Eulerian descent (Worpitzky binning) — none uses Morse–Hedlund counting of distinct contiguous factors. × Deterministic decision-tree depth D(f) of f:{0,1}^n→{0,1} lifted via the XOR-gadget to two-party deterministic communication complexity CC^D(M_{f∘XOR}) on M_f[a,b]=f(a⊕b), in the Raz–McKenzie / Göös–Pitassi–Watson / Hatami–Hosseini–Lovett XOR-lifting framework where CC^D(M_{f∘XOR})=Θ(D(f)·log n) is the canonical Log-Rank Conjecture testbed and the prototypical query-to-communication target.
- **Recorded**: 2026-05-18 20:36 UTC
- **Entry ID**: `1743ebcdf761`

### Statement

Let f:{0,1}^n→{0,1} and let M_f∈{0,1}$^{\{2}$n×2^n} be the XOR-lift M_f[a,b]=f(a⊕b) with b enumerated in standard binary order. For each row r_a, let p_a(n)=#{distinct length-n contiguous substrings of r_a}. Define the averaged scale-n factor complexity SC(f):=(1/2$^{n)\cdot\Sigma_\{a\in\{0,1\}}$n} p_a(n). Then SC(f) ≤ n·2^{D(f)} for every Boolean f, equivalently log_2 SC(f) ≤ D(f)+log_2 n, where D(f) is the deterministic decision-tree depth of f.

### Rationale

A depth-D decision tree partitions {0,1}^n into ≤2^D leaf-cubes, so any row of M_f, read in standard b-order, decomposes into at most 2^D constant-value blocks under coordinate-permutation gauges that fix D coordinates first; a length-n window can straddle at most O(n) block boundaries, capping distinct length-n factors at O(n·2^D). Subword complexity is the canonical 'block count proxy' in symbolic dynamics (Sturmian-shift entropy), exactly the discrete-block quantity that lifting theorems transmute into communication leaves via the Raz–McKenzie gadget composition. The structural form of SC on the lifted matrix gives a non-ring, gauge-fixed proxy for the lifting-induced block-decomposition that bypasses the rank/discrepancy methods previously attempted in this line.

### Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [1810.08668v2] Parity Decision Tree Complexity is Greater Than Granularity - [1506.02936v3] Parity Decision Tree Complexity and 4-Party Communication Complexity of XOR-functions Are Polynomially Equivalent - [1203.2323v2] The greedy flip tree of a subword complex - [1403.8106v1] Recent advances on the log-rank conjecture in communication complexity - [2312.09400v2] Refuting approaches to the log-rank conjecture for XOR functions

**Empirical Test**

- exit code: 1, elapsed: 1.59s

```
99898174, 'instances_tested': 5, 'conjecture_holds': False, 'counterexample': 'n=6, func_type=MAJ_n,
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.7002206645338038, 'instances_tested': 5
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.7055129956660111, 'instances_tested': 5
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.703453628420005, 'instances_tested': 5
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.6975376465407657, 'instances_tested': 5
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.7091381899021894, 'instances_tested': 5
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.7064235731733752, 'instances_tested': 5
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.7005687912699945, 'instances_tested': 5
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.7009164984920413, 'instances_tested': 5
TRIAL: {'metric_name': 'log2_SC(f) - D(f)', 'metric_value': 1.7056269751295425, 'instances_tested': 5

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbo
```

**Judge reasoning**

Safety rail: critic_challenge_falsified | original: A concrete counterexample was produced: for n=6 with MAJ_n, SC(f)=24.71875 exceeds the bound n·2^{D(f)}=24, violating the pre-registered falsification | next: Test whether a relaxed bound SC(f) ≤ c·n·2^{D(f)} (e.g., c=log n or c=2) holds, or whether symmetric functions like MAJ_n systematically saturate/violate the inequality.

---

## Gromov 4-Point Hyperbolicity of Gate-Graph Caps ACC^0[2] Bias on MOD_3

- **Verdict**: INCONCLUSIVE
- **Bridge**: Coarse / Gromov hyperbolic geometry — the 4-point Gromov hyperbolicity δ(G) := (1/2)·sup_{a,b,c,d∈V(G)} [d(a,c)+d(b,d) − max(d(a,b)+d(c,d), d(a,d)+d(b,c))] of a finite undirected graph with hop distance (Gromov 1987 'Hyperbolic groups'; Bridson–Haefliger 1999 'Metric Spaces of Non-Positive Curvature'; Chepoi–Dragan–Estellon–Habib–Vaxes 2008 'Diameters, centers, and approximating trees of δ-hyperbolic geodesic spaces'). Computed in O(|V|^4) by brute-force 4-tuple enumeration over the all-pairs BFS distance matrix; the functional uses max, abs, and integer shortest-path arithmetic with no polynomial F_q extension (shortest-path distance is not preserved under polynomial ring lifts), so the bridge is Aaronson–Wigderson algebrization-safe. The invariant is STRUCTURAL on the circuit DAG (two ACC^0[2] circuits computing the same Boolean function can have very different δ), shielding from Razborov–Rudich natural proofs and avoiding the truth-table Fourier trap that killed Plünnecke doubling on top-Fourier spectra. An arXiv search 'Gromov hyperbolicity' AND ('ACC^0' OR 'Williams algorithmic method' OR 'circuit lower bound' OR 'Boolean circuit') returns 0 direct hits and <5 adjacent papers (network coarse geometry on AS-level Internet graphs; QC analysis on Cayley graphs — never on ACC circuits). Distinct from prior structural circuit attempts: Mostar (edge distance-asymmetry on single edges, not 4-point Gromov product), Forman–Ricci (Bochner edge-vertex local sum), Sinkhorn/Birkhoff (doubly-stochastic spectral scaling), Chang spectrum (Z/p Fourier on DFS labels), additive energy (out-degree L^2), Sidon defect (max representation in F_2^N), fatgraph genus (oriented surface embedding) — none uses the 4-point Gromov product on the gate-graph. × ACC^0[m] circuit size and depth for MOD_q on n inputs with q coprime to m (focus q=3, m=2), in the Williams 2011 / Williams 2014 / Murray–Williams 2018 algorithmic-method regime targeting NEXP ⊄ ACC^0 and explicit-function ACC^0[2] lower bounds. Bridged via the Razborov–Smolensky non-approximability of MOD_3 by low-degree F_2 polynomials: any ACC^0[2] circuit with non-trivial bias toward MOD_3 must wire its gate-graph in a way that

prevents global tree-like (low-δ) coarse geometry, giving a structural obstruction independent of the truth-table side.
- **Recorded**: 2026-05-18 21:12 UTC
- **Entry ID**: `dd0beadbf6d6`

**Statement**

Let C be a connected ACC^0[2] circuit of depth d and gate-count s on n inputs, let G_C be the underlying undirected gate-adjacency graph with hop distance, and let δ(C) := (1/2)·max_{a,b,c,e ∈ V(G_C)} [d(a,c)+d(b,e) − max(d(a,b)+d(c,e), d(a,e)+d(b,c))]. Define the MOD_3 bias B(C) := 2·Pr_{x ∈ {0,1}^n}[C(x) = MOD_3(x)] − 1. Then for every n ≥ 6, $|B(C)|^2 \cdot n \leq (1 + \delta(C)) \cdot 2^{d}$, with a single (C, n) violating this inequality refuting the conjecture.

**Rationale**

Gromov hyperbolicity measures large-scale tree-likeness of finite graphs and is bounded by diameter/2; for constant-depth ACC^0[2] circuits the gate-graph is shallow but its δ can still detect modular obstructions because computing MOD_3 (Razborov–Smolensky-hard for F_2 polynomials) forces the circuit to wire many quasi-isometric 'modular cycles' through MOD_2 gates, raising δ. The 1+δ factor lets tree-like (δ=0) circuits saturate the bound only when $2^d \geq n \cdot |B|^2$, matching the NC^1 regime where balanced binary trees of depth log n can compute MOD_3 exactly while remaining 0-hyperbolic. Because δ depends only on the DAG wiring (not the truth table), the invariant is shielded from natural proofs; because shortest-path distance has no polynomial F_q extension, it is shielded from algebrization.

**Novelty**

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [0906.0693v3] An improved lower bound on the counterfeit coins problem - [2109.03725v3] Quasi-metric antipodal spaces and maximal Gromov hyperbolic spaces - [2407.04826v1] Multi-strategy Based Quantum Cost Reduction of Quantum Boolean Circuits - [1601.02745v1] Basic Reasoning with Tensor Product Representations

**Empirical Test**

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4335b342.py", line 229, in <module>
    trial = run_trial(int(seed))
            ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4335b342.py", line 179, in run_trial
    graph = build_graph(circuit)
            ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4335b342.py", line 147, in build_graph
    node_indices[gate] = index
    ~~~~~~~~~~~~^^^^^^
TypeError: unhashable type: 'list'
```

**Judge reasoning**

The test crashed with a TypeError ('unhashable type: list') in build_graph before any ratios were computed, so no data exists to evaluate the pre-registered support or falsification condition. | next: Fix build_graph to use a hashable node identifier (e.g., gate id or tuple) instead of a list, then rerun the full 14,580-circuit sweep.